PART VI

Extension Fields

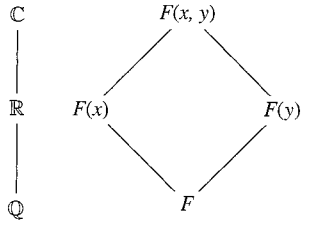
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SECTION 29 INTRODUCTION TO EXTENSION FIELDS

Our Basic Goal Achieved

We are now in a position to achieve our **basic goal,** which, loosely stated, is to show that every nonconstant polynomial has a zero. This will be stated more precisely and proved in Theorem 29.3. We first introduce some new terminology for some old ideas.

29.1 Definition A field E is an **extension field of a field** F if $F \leq E$. ■



29.2 Figure

Thus $\mathbb{R}$ is an extension field of $\mathbb{Q}$, and $\mathbb{C}$ is an extension field of both $\mathbb{R}$ and $\mathbb{Q}$. As in the study of groups, it will often be convenient to use subfield diagrams to picture extension fields, the larger field being on top. We illustrate this in Fig. 29.2. A configuration where there is just one single column of fields, as at the left-hand side of Fig. 29.2, is often referred to, without any precise definition, as a **tower of fields.**

† Section 32 is not required for the remainder of the text.

Now for our *basic goal!* This great and important result follows quickly and elegantly from the techniques we now have at our disposal.

29.3 Theorem **(Kronecker's Theorem) (Basic Goal)** Let F be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Then there exists an extension field E of F and an $\alpha \in E$ such that $f(\alpha) = 0$.

Proof By Theorem 23.20, $f(x)$ has a factorization in $F[x]$ into polynomials that are irreducible over F. Let $p(x)$ be an irreducible polynomial in such a factorization. It is clearly sufficient to find an extension field E of F containing an element α such that $p(\alpha) = 0$.

By Theorem 27.25, $\langle p(x)\rangle$ is a maximal ideal in $F[x]$, so $F[x]/\langle p(x)\rangle$ is a field. We claim that F can be identified with a subfield of $F[x]/\langle p(x)\rangle$ in a natural way by use of the map $\psi : F \to F[x]/\langle p(x)\rangle$ given by

$$\psi(a) = a + \langle p(x)\rangle$$

for $a \in F$. This map is one to one, for if $\psi(a) = \psi(b)$, that is, if $a + \langle p(x)\rangle = b + \langle p(x)\rangle$ for some $a, b \in F$, then $(a - b) \in \langle p(x)\rangle$, so $a - b$ must be a multiple of the polynomial $p(x)$, which is of degree ≥ 1. Now $a, b \in F$ implies that $a - b$ is in F. Thus we must have $a - b = 0$, so $a = b$. We defined addition and multiplication in $F[x]/\langle p(x)\rangle$ by choosing any representatives, so we may choose $a \in (a + \langle p(x)\rangle)$. Thus ψ is a homomorphism that maps F one-to-one onto a subfield of $F[x]/\langle p(x)\rangle$. We identify F with $\{a + \langle p(x)\rangle \mid a \in F\}$ by means of this map ψ. Thus we shall view $E = F[x]/\langle p(x)\rangle$ as an extension field of F. We have now manufactured our desired extension field E of F. It remains for us to show that E contains a zero of $p(x)$.

Let us set

$$\alpha = x + \langle p(x)\rangle,$$

so $\alpha \in E$. Consider the evaluation homomorphism $\phi_\alpha : F[x] \to E$, given by Theorem 22.4. If $p(x) = a_0 + a_1x + \cdots + a_nx^n$, where $a_i \in F$, then we have

$$\phi_\alpha(p(x)) = a_0 + a_1(x + \langle p(x)\rangle) + \cdots + a_n(x + \langle p(x)\rangle)^n$$

Historical Note

Leopold Kronecker is known for his insistence on constructibility of mathematical objects. As he noted, "God made the integers; all else is the work of man." Thus, he wanted to be able to construct new "domains of rationality" (fields) by using only the existence of integers and indeterminates. He did not believe in starting with the real or complex numbers, because as far as he was concerned, those fields could not be determined in a constructive way. Hence in an 1881 paper, Kronecker created an extension field by simply adjoining to a given field a root α of an irreducible nth degree polynomial $p(x)$; that is, his new field consisted of expressions rational in the original field elements and his new root α with the condition that $p(\alpha) = 0$. The proof of the theorem presented in the text (Theorem 29.3) dates from the twentieth century.

Kronecker completed his dissertation in 1845 at the University of Berlin. For many years thereafter, he managed the family business, ultimately becoming financially independent. He then returned to Berlin, where he was elected to the Academy of Sciences and thus permitted to lecture at the university. On the retirement of Kummer, he became a professor at Berlin, and with Karl Weierstrass (1815–1897) directed the influential mathematics seminar.

in $E = F[x]/\langle p(x)\rangle$. *But we can compute in* $F[x]/\langle p(x)\rangle$ *by choosing representatives, and* x *is a representative of the coset* $\alpha = x + \langle p(x)\rangle$. Therefore,

$$\begin{aligned} p(\alpha) &= \left(a_0 + a_1 x + \cdots + a_n x^n\right) + \langle p(x)\rangle \\ &= p(x) + \langle p(x)\rangle = \langle p(x)\rangle = 0 \end{aligned}$$

in $F[x]/\langle p(x)\rangle$. We have found an element α in $E = F[x]/\langle p(x)\rangle$ such that $p(\alpha) = 0$, and therefore $f(\alpha) = 0$. ◆

We illustrate the construction involved in the proof of Theorem 29.3 by two examples.

29.4 Example Let $F = \mathbb{R}$, and let $f(x) = x^2 + 1$, which is well known to have no zeros in $\mathbb{R}$ and thus is irreducible over $\mathbb{R}$ by Theorem 23.10. Then $\langle x^2 + 1\rangle$ is a maximal ideal in $\mathbb{R}[x]$, so $\mathbb{R}[x]/\langle x^2 + 1\rangle$ is a field. Identifying $r \in \mathbb{R}$ with $r + \langle x^2 + 1\rangle$ in $\mathbb{R}[x]/\langle x^2 + 1\rangle$, we can view $\mathbb{R}$ as a subfield of $E = \mathbb{R}[x]/\langle x^2 + 1\rangle$. Let

$$\alpha = x + \langle x^2 + 1\rangle.$$

Computing in $\mathbb{R}[x]/\langle x^2 + 1\rangle$, we find

$$\begin{aligned} \alpha^2 + 1 &= (x + \langle x^2 + 1\rangle)^2 + (1 + \langle x^2 + 1\rangle) \\ &= (x^2 + 1) + \langle x^2 + 1\rangle = 0. \end{aligned}$$

Thus α is a zero of $x^2 + 1$. We shall identify $\mathbb{R}[x]/\langle x^2 + 1\rangle$ with $\mathbb{C}$ at the close of this section. ▲

29.5 Example Let $F = \mathbb{Q}$, and consider $f(x) = x^4 - 5x^2 + 6$. This time $f(x)$ factors in $\mathbb{Q}[x]$ into $(x^2 - 2)(x^2 - 3)$, both factors being irreducible over $\mathbb{Q}$, as we have seen. We can start with $x^2 - 2$ and construct an extension field E of $\mathbb{Q}$ containing α such that $\alpha^2 - 2 = 0$, or we can construct an extension field K of $\mathbb{Q}$ containing an element β such that $\beta^2 - 3 = 0$. The construction in either case is just as in Example 29.4. ▲

Algebraic and Transcendental Elements

As we said before, most of the rest of this text is devoted to the study of zeros of polynomials. We commence this study by putting an element of an extension field E of a field F into one of two categories.

29.6 Definition An element α of an extension field E of a field F is **algebraic over** F if $f(\alpha) = 0$ for some nonzero $f(x) \in F[x]$. If α is not algebraic over F, then α is **transcendental over** F. ■

29.7 Example $\mathbb{C}$ is an extension field of $\mathbb{Q}$. Since $\sqrt{2}$ is a zero of $x^2 - 2$, we see that $\sqrt{2}$ is an algebraic element over $\mathbb{Q}$. Also, i is an algebraic element over $\mathbb{Q}$, being a zero of $x^2 + 1$. ▲

29.8 Example It is well known (but not easy to prove) that the real numbers π and e are transcendental over $\mathbb{Q}$. Here e is the base for the natural logarithms. ▲

Just as we do not speak simply of an *irreducible polynomial,* but rather of an *irreducible polynomial over F*, similarly we don't speak simply of an *algebraic element,* but rather of an *element algebraic over F*. The following illustration shows the reason for this.

29.9 Example The real number π is transcendental over $\mathbb{Q}$, as we stated in Example 29.8. However, π is algebraic over $\mathbb{R}$, for it is a zero of $(x - \pi) \in \mathbb{R}[x]$. ▲

29.10 Example It is easy to see that the real number $\sqrt{1 + \sqrt{3}}$ is algebraic over $\mathbb{Q}$. For if $\alpha = \sqrt{1 + \sqrt{3}}$, then $\alpha^2 = 1 + \sqrt{3}$, so $\alpha^2 - 1 = \sqrt{3}$ and $(\alpha^2 - 1)^2 = 3$. Therefore $\alpha^4 - 2\alpha^2 - 2 = 0$, so α is a zero of $x^4 - 2x^2 - 2$, which is in $\mathbb{Q}[x]$. ▲

To connect these ideas with those of number theory, we give the following definition.

29.11 Definition An element of $\mathbb{C}$ that is algebraic over $\mathbb{Q}$ is an **algebraic number.** A **transcendental number** is an element of $\mathbb{C}$ that is transcendental over $\mathbb{Q}$. ■

There is an extensive and elegant theory of algebraic numbers. (See the Bibliography.)

The next theorem gives a useful characterization of algebraic and transcendental elements over F in an extension field E of F. It also illustrates the importance of our evaluation homomorphisms ϕ_α. *Note that once more we are describing our concepts in terms of mappings.*

29.12 Theorem Let E be an extension field of a field F and let $\alpha \in E$. Let $\phi_\alpha : F[x] \to E$ be the evaluation homomorphism of $F[x]$ into E such that $\phi_\alpha(a) = a$ for $a \in F$ and $\phi_\alpha(x) = \alpha$. Then α is transcendental over F if and only if ϕ_α gives an isomorphism of $F[x]$ with a subdomain of E, that is, if and only if ϕ_α is a one-to-one map.

Proof The element α is transcendental over F if and only if $f(\alpha) \neq 0$ for all nonzero $f(x) \in F[x]$, which is true (by definition) if and only if $\phi_\alpha(f(x)) \neq 0$ for all nonzero $f(x) \in F[x]$, which is true if and only if the kernel of ϕ_α is $\{0\}$, that is, if and only if ϕ_α is a one-to-one map. ◆

The Irreducible Polynomial for α over F

Consider the extension field $\mathbb{R}$ of $\mathbb{Q}$. We know that $\sqrt{2}$ is algebraic over $\mathbb{Q}$, being a zero of $x^2 - 2$. Of course, $\sqrt{2}$ is also a zero of $x^3 - 2x$ and of $x^4 - 3x^2 + 2 = (x^2 - 2)(x^2 - 1)$. Both these other polynomials having $\sqrt{2}$ as a zero were multiples of $x^2 - 2$. The next theorem shows that this is an illustration of a general situation. This theorem plays a central role in our later work.

29.13 Theorem Let E be an extension field of F, and let $\alpha \in E$, where α is algebraic over F. Then there is an irreducible polynomial $p(x) \in F[x]$ such that $p(\alpha) = 0$. This irreducible polynomial $p(x)$ is uniquely determined up to a constant factor in F and is a polynomial of minimal degree ≥ 1 in $F[x]$ having α as a zero. If $f(\alpha) = 0$ for $f(x) \in F[x]$, with $f(x) \neq 0$, then $p(x)$ divides $f(x)$.

Proof Let ϕ_α be the evaluation homomorphism of $F[x]$ into E, given by Theorem 22.4. The kernel of ϕ_α is an ideal and by Theorem 27.24 it must be a principal ideal generated by some $p(x) \in F[x]$. Now $\langle p(x) \rangle$ consists precisely of those elements of $F[x]$ having α as a zero. Thus, if $f(\alpha) = 0$ for $f(x) \neq 0$, then $f(x) \in \langle p(x) \rangle$, so $p(x)$ divides $f(x)$. Thus $p(x)$ is a polynomial of minimal degree ≥ 1 having α as a zero, and any other such polynomial of the same degree as $p(x)$ must be of the form $(a)p(x)$ for some $a \in F$.

It only remains for us to show that $p(x)$ is irreducible. If $p(x) = r(x)s(x)$ were a factorization of $p(x)$ into polynomials of lower degree, then $p(\alpha) = 0$ would imply that $r(\alpha)s(\alpha) = 0$, so either $r(\alpha) = 0$ or $s(\alpha) = 0$, since E is a field. This would contradict the fact that $p(x)$ is of minimal degree ≥ 1 such that $p(\alpha) = 0$. Thus $p(x)$ is irreducible. ◆

By multiplying by a suitable constant in F, we can assume that the coefficient of the highest power of x appearing in $p(x)$ of Theorem 29.13 is 1. Such a polynomial having 1 as the coefficient of the highest power of x appearing is a **monic polynomial.**

29.14 Definition Let E be an extension field of a field F, and let $\alpha \in E$ be algebraic over F. The unique monic polynomial $p(x)$ having the property described in Theorem 29.13 is the **irreducible polynomial for α over F** and will be denoted by $\text{irr}(\alpha, F)$. The degree of $\text{irr}(\alpha, F)$ is the **degree of α over F**, denoted by $\deg(\alpha, F)$. ■

29.15 Example We know that $\text{irr}(\sqrt{2}, \mathbb{Q}) = x^2 - 2$. Referring to Example 29.10, we see that for $\alpha = \sqrt{1 + \sqrt{3}}$ in $\mathbb{R}$, α is a zero of $x^4 - 2x^2 - 2$, which is in $\mathbb{Q}[x]$. Since $x^4 - 2x^2 - 2$ is irreducible over $\mathbb{Q}$ (by Eisenstein with $p = 2$, or by application of the technique of Example 23.14), we see that

$$\text{irr}(\sqrt{1 + \sqrt{3}}, \mathbb{Q}) = x^4 - 2x^2 - 2.$$

Thus $\sqrt{1 + \sqrt{3}}$ is algebraic of degree 4 over $\mathbb{Q}$. ▲

Just as we must speak of an element α as *algebraic over F* rather than simply as *algebraic,* we must speak of the *degree of α over F* rather than the *degree of α*. To take a trivial illustration, $\sqrt{2} \in \mathbb{R}$ is algebraic of degree 2 over $\mathbb{Q}$ but algebraic of degree 1 over $\mathbb{R}$, for $\text{irr}(\sqrt{2}, \mathbb{R}) = x - \sqrt{2}$.

The quick development of the theory here is due to the machinery of homomorphisms and ideal theory that we now have at our disposal. Note especially our constant use of the evaluation homomorphisms ϕ_α.

Simple Extensions

Let E be an extension field of a field F, and let $\alpha \in E$. Let ϕ_α be the evaluation homomorphism of $F[x]$ into E with $\phi_\alpha(a) = a$ for $a \in F$ and $\phi_\alpha(x) = \alpha$, as in Theorem 22.4. We consider two cases.

Case I *Suppose α is algebraic over F.* Then as in Theorem 29.13, the kernel of ϕ_α is $\langle \text{irr}(\alpha, F)\rangle$ and by Theorem 27.25, $\langle \text{irr}(\alpha, F)\rangle$ is a maximal ideal of $F[x]$. Therefore, $F[x]/\langle \text{irr}(\alpha, F)\rangle$ is a field and is isomorphic to the image $\phi_\alpha[F[x]]$ in E. This subfield $\phi_\alpha[F[x]]$ of E is then the smallest subfield of E containing F and α. We shall denote this field by $F(\alpha)$.

Case II *Suppose α is transcendental over F.* Then by Theorem 29.12, ϕ_α gives an isomorphism of $F[x]$ with a subdomain of E. Thus in this case $\phi_\alpha[F[x]]$ is *not* a field but an integral domain that we shall denote by $F[\alpha]$. By Corollary 21.8, E contains a field of quotients of $F[\alpha]$, which is thus the smallest subfield of E containing F and α. As in Case I, we denote this field by $F(\alpha)$.

29.16 Example Since π is transcendental over $\mathbb{Q}$, the field $\mathbb{Q}(\pi)$ is isomorphic to the field $\mathbb{Q}(x)$ of rational functions over $\mathbb{Q}$ in the indeterminate x. Thus from a structural viewpoint, an element that is transcendental over a field F behaves as though it were an indeterminate over F. ▲

29.17 Definition An extension field E of a field F is a **simple extension of** F if $E = F(\alpha)$ for some $\alpha \in E$. ■

Many important results appear throughout this section. We have now developed so much machinery that results are starting to pour out of our efficient plant at an alarming rate. The next theorem gives us insight into the nature of the field $F(\alpha)$ in the case where α is algebraic over F.

29.18 Theorem Let E be a simple extension $F(\alpha)$ of a field F, and let α be algebraic over F. Let the degree of $\text{irr}(\alpha, F)$ be $n \geq 1$. Then every element β of $E = F(\alpha)$ can be uniquely expressed in the form

$$\beta = b_0 + b_1\alpha + \cdots + b_{n-1}\alpha^{n-1},$$

where the b_i are in F.

Proof For the usual evaluation homomorphism ϕ_α, every element of

$$F(\alpha) = \phi_\alpha[F[x]]$$

is of the form $\phi_\alpha(f(x)) = f(\alpha)$, a formal polynomial in α with coefficients in F. Let

$$\text{irr}(\alpha, F) = p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0.$$

Then $p(\alpha) = 0$, so

$$\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_0.$$

This equation in $F(\alpha)$ can be used to express every monomial α^m for $m \geq n$ in terms of powers of α that are less than n. For example,

$$\alpha^{n+1} = \alpha\alpha^n = -a_{n-1}\alpha^n - a_{n-2}\alpha^{n-1} - \cdots - a_0\alpha$$
$$= -a_{n-1}(-a_{n-1}\alpha^{n-1} - \cdots - a_0) - a_{n-2}\alpha^{n-1} - \cdots - a_0\alpha.$$

Thus, if $\beta \in F(\alpha)$, β can be expressed in the required form

$$\beta = b_0 + b_1\alpha + \cdots + b_{n-1}\alpha^{n-1}.$$

For uniqueness, if

$$b_0 + b_1\alpha + \cdots + b_{n-1}\alpha^{n-1} = b_0' + b_1'\alpha + \cdots + b_{n-1}'\alpha^{n-1}$$

for $b_i' \in F$, then

$$(b_0 - b_0') + (b_1 - b_1')x + \cdots + (b_{n-1} - b_{n-1}')x^{n-1} = g(x)$$

is in $F[x]$ and $g(\alpha) = 0$. Also, the degree of $g(x)$ is less than the degree of $\text{irr}(\alpha, F)$. Since $\text{irr}(\alpha, F)$ is a nonzero polynomial of minimal degree in $F[x]$ having α as a zero, we must have $g(x) = 0$. Therefore, $b_i - b_i' = 0$, so

$$b_i = b_i',$$

and the uniqueness of the b_i is established. ◆

We give an impressive example illustrating Theorem 29.18.

29.19 Example The polynomial $p(x) = x^2 + x + 1$ in $\mathbb{Z}_2[x]$ is irreducible over $\mathbb{Z}_2$ by Theorem 23.10, since neither element 0 nor element 1 of $\mathbb{Z}_2$ is a zero of $p(x)$. By Theorem 29.3, we know that there is an extension field E of $\mathbb{Z}_2$ containing a zero α of $x^2 + x + 1$. By Theorem 29.18, $\mathbb{Z}_2(\alpha)$ has as elements $0 + 0\alpha$, $1 + 0\alpha$, $0 + 1\alpha$, and $1 + 1\alpha$, that is, 0, 1, α, and $1 + \alpha$. *This gives us a new finite field, of four elements!* The addition and multiplication tables for this field are shown in Tables 29.20 and 29.21. For example, to compute $(1 + \alpha)(1 + \alpha)$ in $\mathbb{Z}_2(\alpha)$, we observe that since $p(\alpha) = \alpha^2 + \alpha + 1 = 0$, then

$$\alpha^2 = -\alpha - 1 = \alpha + 1.$$

Therefore,

$$(1 + \alpha)(1 + \alpha) = 1 + \alpha + \alpha + \alpha^2 = 1 + \alpha^2 = 1 + \alpha + 1 = \alpha.$$ ▲

Finally, we can use Theorem 29.18 to fulfill our promise of Example 29.4 and show that $\mathbb{R}[x]/\langle x^2 + 1\rangle$ is isomorphic to the field $\mathbb{C}$ of complex numbers. We saw in Example 29.4 that we can view $\mathbb{R}[x]/\langle x^2 + 1\rangle$ as an extension field of $\mathbb{R}$. Let

$$\alpha = x + \langle x^2 + 1\rangle.$$

29.20 Table

$+$	0	1	α	$1+\alpha$
0	0	1	α	$1+\alpha$
1	1	0	$1+\alpha$	α
α	α	$1+\alpha$	0	1
$1+\alpha$	$1+\alpha$	α	1	0

29.21 Table

	0	1	α	$1+\alpha$
0	0	0	0	0
1	0	1	α	$1+\alpha$
α	0	α	$1+\alpha$	1
$1+\alpha$	0	$1+\alpha$	1	α

Then $\mathbb{R}(\alpha) = \mathbb{R}[x]/\langle x^2 + 1\rangle$ and consists of all elements of the form $a + b\alpha$ for $a, b \in \mathbb{R}$, by Theorem 29.18. But since $\alpha^2 + 1 = 0$, we see that α plays the role of $i \in \mathbb{C}$, and $a + b\alpha$ plays the role of $(a + bi) \in \mathbb{C}$. Thus $\mathbb{R}(\alpha) \simeq \mathbb{C}$. *This is the elegant algebraic way to construct* $\mathbb{C}$ *from* $\mathbb{R}$.

EXERCISES 29

Computations

In Exercises 1 through 5, show that the given number $\alpha \in \mathbb{C}$ is algebraic over $\mathbb{Q}$ by finding $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$.

1. $1 + \sqrt{2}$ **2.** $\sqrt{2} + \sqrt{3}$ **3.** $1 + i$

4. $\sqrt{1 + \sqrt[3]{2}}$ **5.** $\sqrt{\sqrt[3]{2} - i}$

In Exercises 6 through 8, find $\text{irr}(\alpha, \mathbb{Q})$ and $\deg(\alpha, \mathbb{Q})$ for the given algebraic number $\alpha \in \mathbb{C}$. Be prepared to prove that your polynomials are irreducible over $\mathbb{Q}$ if challenged to do so.

6. $\sqrt{3 - \sqrt{6}}$ **7.** $\sqrt{(\frac{1}{3}) + \sqrt{7}}$ **8.** $\sqrt{2} + i$

In Exercises 9 through 16, classify the given $\alpha \in \mathbb{C}$ as algebraic or transcendental over the given field F. If α is algebraic over F, find $\deg(\alpha, F)$.

9. $\alpha = i, F = \mathbb{Q}$ **10.** $\alpha = 1 + i, F = \mathbb{R}$

11. $\alpha = \sqrt{\pi}, F = \mathbb{Q}$ **12.** $\alpha = \sqrt{\pi}, F = \mathbb{R}$

13. $\alpha = \sqrt{\pi}, F = \mathbb{Q}(\pi)$ **14.** $\alpha = \pi^2, F = \mathbb{Q}$

15. $\alpha = \pi^2, F = \mathbb{Q}(\pi)$ **16.** $\alpha = \pi^2, F = \mathbb{Q}(\pi^3)$

17. Refer to Example 29.19 of the text. The polynomial $x^2 + x + 1$ has a zero α in $\mathbb{Z}_2(\alpha)$ and thus must factor into a product of linear factors in $(\mathbb{Z}_2(\alpha))[x]$. Find this factorization. [*Hint:* Divide $x^2 + x + 1$ by $x - \alpha$ by long division, using the fact that $\alpha^2 = \alpha + 1$.]

18. **a.** Show that the polynomial $x^2 + 1$ is irreducible in $\mathbb{Z}_3[x]$.

b. Let α be a zero of $x^2 + 1$ in an extension field of $\mathbb{Z}_3$. As in Example 29.19, give the multiplication and addition tables for the nine elements of $\mathbb{Z}_3(\alpha)$, written in the order $0, 1, 2, \alpha, 2\alpha, 1 + \alpha, 1 + 2\alpha, 2 + \alpha$, and $2 + 2\alpha$.

Concepts

In Exercises 19 through 22, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

19. An element α of an extension field E of a field F if *algebraic over* F if and only if α is a zero of some polynomial.

20. An element β of an extension field E of a field F is *transcendental over* F if and only if β is not a zero of any polynomial in $F[x]$.

21. A *monic polynomial in* $F[x]$ is one having all coefficients equal to 1.

22. A field E is a *simple extension* of a subfield F if and only if there exists some $\alpha \in E$ such that no proper subfield of E contains α.

23. Mark each of the following true or false.

_____ **a.** The number π is transcendental over $\mathbb{Q}$.
_____ **b.** $\mathbb{C}$ is a simple extension of $\mathbb{R}$.
_____ **c.** Every element of a field F is algebraic over F.
_____ **d.** $\mathbb{R}$ is an extension field of $\mathbb{Q}$.
_____ **e.** $\mathbb{Q}$ is an extension field of $\mathbb{Z}_2$.
_____ **f.** Let $\alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$ of degree n. If $f(\alpha) = 0$ for nonzero $f(x) \in \mathbb{Q}[x]$, then (degree $f(x)) \geq n$.
_____ **g.** Let $\alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$ of degree n. If $f(\alpha) = 0$ for nonzero $f(x) \in \mathbb{R}[x]$, then (degree $f(x)) \geq n$.
_____ **h.** Every nonconstant polynomial in $F[x]$ has a zero in some extension field of F.
_____ **i.** Every nonconstant polynomial in $F[x]$ has a zero in every extension field of F.
_____ **j.** If x is an indeterminate, $\mathbb{Q}[\pi] \simeq \mathbb{Q}[x]$.

24. We have stated without proof that π and e are transcendental over $\mathbb{Q}$.

a. Find a subfield F of $\mathbb{R}$ such that π is algebraic of degree 3 over F.
b. Find a subfield E of $\mathbb{R}$ such that e^2 is algebraic of degree 5 over E.

25. **a.** Show that $x^3 + x^2 + 1$ is irreducible over $\mathbb{Z}_2$.
b. Let α be a zero of $x^3 + x^2 + 1$ in an extension field of $\mathbb{Z}_2$. Show that $x^3 + x^2 + 1$ factors into three linear factors in $(\mathbb{Z}_2(\alpha))[x]$ by actually finding this factorization. [*Hint:* Every element of $\mathbb{Z}_2(\alpha)$ is of the form

$$a_0 + a_1\alpha + a_2\alpha^2 \quad \text{for} \quad a_i = 0, 1.$$

Divide $x^3 + x^2 + 1$ by $x - \alpha$ by long division. Show that the quotient also has a zero in $\mathbb{Z}_2(\alpha)$ by simply trying the eight possible elements. Then complete the factorization.]

26. Let E be an extension field of $\mathbb{Z}_2$ and let $\alpha \in E$ be algebraic of degree 3 over $\mathbb{Z}_2$. Classify the groups $\langle \mathbb{Z}_2(\alpha), + \rangle$ and $\langle (\mathbb{Z}_2(\alpha))^*, \cdot \rangle$ according to the Fundamental Theorem of finitely generated abelian groups. As usual, $(\mathbb{Z}_2(\alpha))^*$ is the set of nonzero elements of $\mathbb{Z}_2(\alpha)$.

27. Let E be an extension field of a field F and let $\alpha \in E$ be algebraic over F. The polynomial irr(α, F) is sometimes referred to as the **minimal polynomial for α over F**. Why is this designation appropriate?

Proof Synopsis

28. Give a two- or three-sentence synopsis of Theorem 29.3.

Theory

29. Let E be an extension field of F, and let $\alpha, \beta \in E$. Suppose α is transcendental over F but algebraic over $F(\beta)$. Show that β is algebraic over $F(\alpha)$.

30. Let E be an extension field of a finite field F, where F has q elements. Let $\alpha \in E$ be algebraic over F of degree n. Prove that $F(\alpha)$ has q^n elements.

31. **a.** Show that there exists an irreducible polynomial of degree 3 in $\mathbb{Z}_3[x]$.
b. Show from part (a) that there exists a finite field of 27 elements. [*Hint:* Use Exercise 30.]

32. Consider the prime field $\mathbb{Z}_p$ of characteristic $p \neq 0$.

a. Show that, for $p \neq 2$, not every element in $\mathbb{Z}_p$ is a square of an element of $\mathbb{Z}_p$. [*Hint:* $1^2 = (p-1)^2 = 1$ in $\mathbb{Z}_p$. Deduce the desired conclusion *by counting*.]

b. Using part (a), show that there exist finite fields of p^2 elements for every prime p in $\mathbb{Z}^+$.

33. Let E be an extension field of a field F and let $\alpha \in E$ be transcendental over F. Show that every element of $F(\alpha)$ that is not in F is also transcendental over F.

34. Show that $\{a + b(\sqrt[3]{2}) + c(\sqrt[3]{2})^2 \mid a, b, c \in \mathbb{Q}\}$ is a subfield of $\mathbb{R}$ by using the ideas of this section, rather than by a formal verification of the field axioms. [*Hint:* Use Theorem 29.18.]

35. Following the idea of Exercise 31, show that there exists a field of 8 elements; of 16 elements; of 25 elements.

36. Let F be a finite field of characteristic p. Show that every element of F is algebraic over the prime field $\mathbb{Z}_p \leq F$. [*Hint:* Let F^* be the set of nonzero elements of F. Apply group theory to the group $\langle F^*, \cdot \rangle$ to show that every $\alpha \in F^*$ is a zero of some polynomial in $\mathbb{Z}_p[x]$ of the form $x^n - 1$.]

37. Use Exercises 30 and 36 to show that every finite field is of prime-power order, that is, it has a prime-power number of elements.

SECTION 30 VECTOR SPACES

The notions of a vector space, scalars, independent vectors, and bases may be familiar. In this section, we present these ideas where the scalars may be elements of any field. We use Greek letters like α and β for vectors since, in our application, the vectors will be elements of an extension field E of a field F. The proofs are all identical with those often given in a first course in linear algebra. If these ideas are familiar, we suggest studying Examples 30.4, 30.8, 30.11, 30.14, and 30.22, and then reading Theorem 30.23 and its proof. If the examples and the theorem are understood, then do some exercises and proceed to the next section.

Definition and Elementary Properties

The topic of vector spaces is the cornerstone of linear algebra. Since linear algebra is not the subject for study in this text, our treatment of vector spaces will be brief, designed to develop only the concepts of linear independence and dimension that we need for our field theory.

The terms *vector* and *scalar* are probably familiar from calculus. Here we allow scalars to be elements of any field, not just the real numbers, and develop the theory by axioms just as for the other algebraic structures we have studied.

30.1 Definition Let F be a field. A **vector space over** F (or F**-vector space**) consists of an abelian group V under addition together with an operation of scalar multiplication of each element of V by each element of F on the left, such that for all $a, b \in F$ and $\alpha, \beta \in V$ the following

conditions are satisfied:

$\mathscr{V}_1$. $a\alpha \in V$.
$\mathscr{V}_2$. $a(b\alpha) = (ab)\alpha$.
$\mathscr{V}_3$. $(a + b)\alpha = (a\alpha) + (b\alpha)$.
$\mathscr{V}_4$. $a(\alpha + \beta) = (a\alpha) + (a\beta)$.
$\mathscr{V}_5$. $1\alpha = \alpha$.

The elements of V are **vectors** and the elements of F are **scalars.** When only one field F is under discussion, we drop the reference to F and refer to a *vector space.* ■

Note that scalar multiplication for a vector space is not a binary operation on one set in the sense we defined it in Section 2. It associates an element $a\alpha$ of V with each ordered pair (a, α), consisting of an element a of F and an element α of V. Thus scalar multiplication is a *function* mapping $F \times V$ into V. Both the additive identity for V, the 0-vector, and the additive identity for F, the 0-scalar, will be denoted by 0.

30.2 Example Consider the abelian group $\langle \mathbb{R}_n, + \rangle = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ for n factors, which consists of ordered n-tuples under addition by components. Define scalar multiplication for scalars in $\mathbb{R}$ by

$$r\alpha = (ra_1, \cdots, ra_n)$$

■ Historical Note

The ideas behind the abstract notion of a vector space occurred in many concrete examples during the nineteenth century and earlier. For example, William Rowan Hamilton dealt with complex numbers explicitly as pairs of real numbers and, as noted in Section 24, also dealt with triples and eventually quadruples of real numbers in his invention of the quaternions. In these cases, the "vectors" turned out to be objects which could both be added and multiplied by scalars, using "reasonable" rules for both of these operations. Other examples of such objects included differential forms (expressions under integral signs) and algebraic integers.

Although Hermann Grassmann (1809–1877) succeeded in working out a detailed theory of n-dimensional spaces in his *Die Lineale Ausdehnungslehre* of 1844 and 1862, the first mathematician to give an abstract definition of a vector space equivalent to Definition 30.1 was Giuseppe Peano (1858–1932) in his *Calcolo Geometrico* of 1888. Peano's aim in the book, as the title indicates, was to develop a geometric calculus. According to Peano, such a calculus "consists of a system of operations analogous to those of algebraic calculus, but in which the objects with which the calculations are performed are, instead of numbers, geometrical objects." Curiously, Peano's work had no immediate effect on the mathematical scene. Although Hermann Weyl (1885–1955) essentially repeated Peano's definition in his *Space-Time-Matter* of 1918, the definition of a vector space did not enter the mathematical mainstream until it was announced for a third time by Stefan Banach (1892–1945) in the 1922 publication of his dissertation dealing with what we now call *Banach spaces,* complete normed vector spaces.

for $r \in \mathbb{R}$ and $\alpha = (a_1, \cdots, a_n) \in \mathbb{R}^n$. With these operations, $\mathbb{R}^n$ becomes a vector space over $\mathbb{R}$. The axioms for a vector space are readily checked. In particular, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ as a vector space over $\mathbb{R}$ can be viewed as all "vectors whose starting points are the origin of the Euclidean plane" in the sense often studied in calculus courses. ▲

30.3 Example For any field F, $F[x]$ can be viewed as a vector space over F, where addition of vectors is ordinary addition of polynomials in $F[x]$ and scalar multiplication $a\alpha$ of an element of $F[x]$ by an element of F is ordinary multiplication in $F[x]$. The axioms $\mathscr{V}_1$ through $\mathscr{V}_5$ for a vector space then follow immediately from the fact that $F[x]$ is a ring with unity. ▲

30.4 Example Let E be an extension field of a field F. Then E can be regarded as a vector space over F, where addition of vectors is the usual addition in E and scalar multiplication $a\alpha$ is the usual field multiplication in E with $a \in F$ and $\alpha \in E$. The axioms follow at once from the field axioms for E. Here our field of scalars is actually a subset of our space of vectors. *It is this example that is the important one for us.* ▲

We are assuming nothing about vector spaces from previous work and shall prove everything we need from the definition, even though the results may be familiar from calculus.

30.5 Theorem If V is a vector space over F, then $0\alpha = 0$, $a0 = 0$ and $(-a)\alpha = a(-\alpha) = -(a\alpha)$ for all $a \in F$ and $\alpha \in V$.

Proof The equation $0\alpha = 0$ is to be read "(0-scalar)α = 0-vector." Likewise, $a0 = 0$ is to be read "a(0-vector) = 0-vector." The proofs here are very similar to those in Theorem 18.8 for a ring and again depend heavily on the distributive laws $\mathscr{V}_3$ and $\mathscr{V}_4$. Now

$$(0\alpha) = (0 + 0)\alpha = (0\alpha) + (0\alpha)$$

is an equation in the abelian group $\langle V, + \rangle$, so by the group cancellation law, $0 = 0\alpha$. Likewise, from

$$a0 = a(0 + 0) = a0 + a0,$$

we conclude that $a0 = 0$. Then

$$0 = 0\alpha = (a + (-a))\alpha = a\alpha + (-a)\alpha,$$

so $(-a)\alpha = -(a\alpha)$. Likewise, from

$$0 = a0 = a(\alpha + (-\alpha)) = a\alpha + a(-\alpha),$$

we conclude that $a(-\alpha) = -(a\alpha)$ also. ◆

Linear Independence and Bases

30.6 Definition Let V be a vector space over F. The vectors in a subset $S = \{\alpha_i \mid i \in I\}$ of V **span** (or **generate**) V if for every $\beta \in V$, we have

$$\beta = a_1\alpha_{i_1} + a_2\alpha_{i_2} + \cdots + a_n\alpha_{i_n}$$

for some $a_j \in F$ and $\alpha_{i_j} \in S$, $j = 1, \cdots, n$. A vector $\sum_{j=1}^{n} a_j\alpha_{i_j}$ is a **linear combination of the** α_{i_j}. ■

30.7 Example In the vector space $\mathbb{R}^n$ over $\mathbb{R}$ of Example 30.2, the vectors

$$(1, 0, \cdots, 0), (0, 1, \cdots, 0), \cdots, (0, 0, \cdots, 1)$$

clearly span $\mathbb{R}^n$, for

$$(a_1, a_2, \cdots, a_n) = a_1(1, 0, \cdots, 0) + a_2(0, 1, \cdots, 0) + \cdots + a_n(0, 0, \cdots, 1).$$

Also, the monomials x^m for $m \geq 0$ span $F[x]$ over F, the vector space of Example 30.3. ▲

30.8 Example Let F be a field and E an extension field of F. Let $\alpha \in E$ be algebraic over F. Then $F(\alpha)$ is a vector space over F and by Theorem 30.18, it is spanned by the vectors in $\{1, \alpha, \cdots, \alpha^{n-1}\}$, where $n = \deg(\alpha, F)$. *This is the important example for us.* ▲

30.9 Definition A vector space V over a field F is **finite dimensional** if there is a finite subset of V whose vectors span V. ■

30.10 Example Example 30.7 shows that $\mathbb{R}^n$ is finite dimensional. The vector space $F[x]$ over F is *not* finite dimensional, since polynomials of arbitrarily large degree could not be linear combinations of elements of any *finite* set of polynomials. ▲

30.11 Example If $F \leq E$ and $\alpha \in E$ is algebraic over the field F, Example 30.8 shows that $F(\alpha)$ is a finite-dimensional vector space over F. *This is the most important example for us.* ▲

The next definition contains the most important idea in this section.

30.12 Definition The vectors in a subset $S = \{\alpha_i \mid i \in I\}$ of a vector space V over a field F are **linearly independent over** F if, for any distinct vectors $\alpha_{i_j} \in S$, coefficients $a_j \in F$ and $n \in \mathbb{Z}^+$, we have $\sum_{j=1}^{n} a_j \alpha_{i_j} = 0$ in V only if $a_j = 0$ for $j = 1, \cdots, n$. If the vectors are not linearly independent over F, they are **linearly dependent over** F. ■

Thus the vectors in $\{\alpha_i \mid i \in I\}$ are linearly independent over F if the only way the 0-vector can be expressed as a linear combination of the vectors α_i is to have all scalar coefficients equal to 0. If the vectors are linearly dependent over F, then there exist $a_j \in F$ for $j = 1, \cdots, n$ such that $\sum_{j=1}^{n} a_j \alpha_{i_j} = 0$, where not all $a_j = 0$.

30.13 Example Observe that the vectors spanning the space $\mathbb{R}^n$ that are given in Example 30.7 are linearly independent over $\mathbb{R}$. Likewise, the vectors in $\{x^m \mid m \geq 0\}$ are linearly independent vectors of $F[x]$ over F. Note that $(1, -1)$, $(2, 1)$, and $(-3, 2)$ are linearly dependent in $\mathbb{R}^2$ over $\mathbb{R}$, since

$$7(1, -1) + (2, 1) + 3(-3, 2) = (0, 0) = 0.$$ ▲

30.14 Example Let E be an extension field of a field F, and let $\alpha \in E$ be algebraic over F. If $\deg(\alpha, F) = n$, then by Theorem 29.18, every element of $F(\alpha)$ can be *uniquely* expressed in the form

$$b_0 + b_1\alpha + \cdots + b_{n-1}\alpha^{n-1}$$

for $b_i \in F$. In particular, $0 = 0 + 0\alpha + \cdots + 0\alpha^{n-1}$ must be a *unique* such expression for 0. Thus the elements $1, \alpha, \cdots, \alpha^{n-1}$ are linearly independent vectors in $F(\alpha)$ over

the field F. They also span $F(\alpha)$, so by the next definition, $1, \alpha, \cdots, \alpha^{n-1}$ form a *basis* for $F(\alpha)$ over F. *This is the important example for us.* In fact, this is the reason we are doing this material on vector spaces. ▲

30.15 Definition If V is a vector space over a field F, the vectors in a subset $B = \{\beta_i \mid i \in I\}$ of V form a **basis for V over F** if they span V and are linearly independent. ■

Dimension

The only other results we wish to prove about vector spaces are that every finite-dimensional vector space has a basis, and that any two bases of a finite-dimensional vector space have the same number of elements. Both these facts are true without the assumption that the vector space is finite dimensional, but the proofs require more knowledge of set theory than we are assuming, and the finite-dimensional case is all we need. First we give an easy lemma.

30.16 Lemma Let V be a vector space over a field F, and let $\alpha \in V$. If α is a linear combination of vectors β_i in V for $i = 1, \cdots, m$ and each β_i is a linear combination of vectors γ_j in V for $j = 1, \cdots, n$, then α is a linear combination of the γ_j.

Proof Let $\alpha = \sum_{i=1}^{m} a_i \beta_i$, and let $\beta_i = \sum_{j=1}^{n} b_{ij} \gamma_j$, where a_i and b_{ij} are in F. Then

$$\alpha = \sum_{i=1}^{m} a_i \left(\sum_{j=1}^{n} b_{ij} \gamma_j \right) = \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_i b_{ij} \right) \gamma_j,$$

and $(\sum_{i=1}^{m} a_i b_{ij}) \in F$. ◆

30.17 Theorem In a finite-dimensional vector space, every finite set of vectors spanning the space contains a subset that is a basis.

Proof Let V be finite dimensional over F, and let vectors $\alpha_1, \cdots, \alpha_n$ in V span V. Let us list the α_i in a row. Examine each α_i in succession, starting at the left with $i = 1$, and discard the first α_j that is some linear combination of the preceding α_i for $i < j$. Then continue, starting with the following α_{j+1}, and discard the next α_k that is some linear combination of its remaining predecessors, and so on. When we reach α_n after a finite number of steps, those α_i remaining in our list are such that none is a linear combination of the preceding α_i in this reduced list. Lemma 30.16 shows that any vector that is a linear combination of the original collection of α_i is still a linear combination of our reduced, and possibly smaller, set in which no α_i is a linear combination of its predecessors. Thus the vectors in the reduced set of α_i again span V.

For the reduced set, suppose that

$$a_1 \alpha_{i_1} + \cdots + a_r \alpha_{i_r} = 0$$

for $i_1 < i_2 < \cdots < i_r$ and that some $a_j \neq 0$. We may assume from Theorem 30.5 that $a_r \neq 0$, or we could drop $a_r \alpha_{i_r}$ from the left side of the equation. Then, using

Theorem 30.5 again, we obtain

$$\alpha_{i_r} = \left(-\frac{a_1}{a_r}\right)\alpha_{i_1} + \cdots + \left(-\frac{a_{r-1}}{a_r}\right)\alpha_{i_{r-1}},$$

which shows that α_{i_r} is a linear combination of its predecessors, contradicting our construction. Thus the vectors α_i in the reduced set both span V and are linearly independent, so they form a basis for V over F. ◆

30.18 Corollary A finite-dimensional vector space has a finite basis.

Proof By definition, a finite-dimensional vector space has a finite set of vectors that span the space. Theorem 30.17 completes the proof. ◆

The next theorem is the culmination of our work on vector spaces.

30.19 Theorem Let $S = \{\alpha_1, \cdots, \alpha_r\}$ be a finite set of linearly independent vectors of a finite-dimensional vector space V over a field F. Then S can be enlarged to a basis for V over F. Furthermore, if $B = \{\beta_1, \cdots, \beta_n\}$ is any basis for V over F, then $r \leq n$.

Proof By Corollary 30.18, there is a basis $B = \{\beta_1, \cdots, \beta_n\}$ for V over F. Consider the finite sequence of vectors

$$\alpha_1, \cdots, \alpha_r, \beta_1, \cdots, \beta_n.$$

These vectors span V, since B is a basis. Following the technique, used in Theorem 30.17, of discarding in turn each vector that is a linear combination of its remaining predecessors, working from left to right, we arrive at a basis for V. Observe that no α_i is cast out, since the α_i are linearly independent. Thus S can be enlarged to a basis for V over F.

For the second part of the conclusion, consider the sequence

$$\alpha_1, \beta_1, \cdots, \beta_n.$$

These vectors are not linearly independent over F, because α_1 is a linear combination

$$\alpha_1 = b_1\beta_1 + \cdots + b_n\beta_n,$$

since the β_i form a basis. Thus

$$\alpha_1 + (-b_1)\beta_1 + \cdots + (-b_n)\beta_n = 0.$$

The vectors in the sequence do span V, and if we form a basis by the technique of working from left to right and casting out in turn each vector that is a linear combination of its remaining predecessors, at least one β_i must be cast out, giving a basis

$$\{\alpha_1, \beta_1^{(1)}, \cdots, \beta_m^{(1)}\},$$

where $m \leq n - 1$. Applying the same technique to the sequence of vectors

$$\alpha_1, \alpha_2, \beta_1^{(1)}, \cdots, \beta_m^{(1)},$$

we arrive at a new basis

$$\{\alpha_1, \alpha_2, \beta_1^{(2)}, \cdots, \beta_s^{(2)}\},$$

with $s \leq n-2$. Continuing, we arrive finally at a basis

$$\{\alpha_1, \cdots, \alpha_r, \beta_1^{(r)}, \cdots, \beta_t^{(r)}\},$$

where $0 \leq t \leq n-r$. Thus $r \leq n$. ◆

30.20 Corollary Any two bases of a finite-dimensional vector space V over F have the same number of elements.

Proof Let $B = \{\beta_1, \cdots, \beta_n\}$ and $B' = \{\beta_1', \cdots, \beta_m'\}$ be two bases. Then by Theorem 30.19, regarding B as an independent set of vectors and B' as a basis, we see that $n \leq m$. A symmetric argument gives $m \leq n$, so $m = n$. ◆

30.21 Definition If V is a finite-dimensional vector space over a field F, the number of elements in a basis (independent of the choice of basis, as just shown) is the **dimension of V over F.** ■

30.22 Example Let E be an extension field of a field F, and let $\alpha \in E$. Example 30.14 shows that if α is algebraic over F and $\deg(\alpha, F) = n$, then the dimension of $F(\alpha)$ as a vector space over F is n. *This is the important example for us.* ▲

An Application to Field Theory

We collect the results of field theory contained in Examples 30.4, 30.8, 30.11, 30.14, and 30.22, and incorporate them into one theorem. The last sentence of this theorem gives an additional nice application of these vector space ideas to field theory.

30.23 Theorem Let E be an extension field of F, and let $\alpha \in E$ be algebraic over F. If $\deg(\alpha, F) = n$, then $F(\alpha)$ is an n-dimensional vector space over F with basis $\{1, \alpha, \cdots, \alpha^{n-1}\}$. Furthermore, every element β of $F(\alpha)$ is algebraic over F, and $\deg(\beta, F) \leq \deg(\alpha, F)$.

Proof We have shown everything in the preceding examples except the very important result stated in the last sentence of the above theorem. Let $\beta \in F(\alpha)$, where α is algebraic over F of degree n. Consider the elements

$$1, \beta, \beta^2, \cdots, \beta^n.$$

These cannot be $n+1$ distinct elements of $F(\alpha)$ that are linearly independent over F, for by Theorem 30.19, any basis of $F(\alpha)$ over F would have to contain at least as many elements as are in any set of linearly independent vectors over F. However, the basis $\{1, \alpha, \cdots, \alpha^{n-1}\}$ has just n elements. If $\beta^i = \beta^j$, then $\beta^i - \beta^j = 0$, so in any case there exist $b_i \in F$ such that

$$b_0 + b_1\beta + b_2\beta^2 + \cdots + b_n\beta^n = 0,$$

where not all $b_i = 0$. Then $f(x) = b_n x^n + \cdots + b_1 x + b_0$ is a nonzero element of $F[x]$ such that $f(\beta) = 0$. Therefore, β is algebraic over F and $\deg(\beta, F)$ is at most n. ◆

EXERCISES 30

Computations

1. Find three bases for $\mathbb{R}^2$ over $\mathbb{R}$, no two of which have a vector in common.

In Exercises 2 and 3, determine whether the given set of vectors is a basis for $\mathbb{R}^3$ over $\mathbb{R}$.

2. $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$

3. $\{(-1, 1, 2), (2, -3, 1), (10, -14, 0)\}$

In Exercises 4 through 9, give a basis for the indicated vector space over the field.

4. $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$

5. $\mathbb{R}(\sqrt{2})$ over $\mathbb{R}$

6. $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$

7. $\mathbb{C}$ over $\mathbb{R}$

8. $\mathbb{Q}(i)$ over $\mathbb{Q}$

9. $\mathbb{Q}(\sqrt[4]{2})$ over $\mathbb{Q}$

10. According to Theorem 30.23, the element $1 + \alpha$ of $\mathbb{Z}_2(\alpha)$ of Example 29.19 is algebraic over $\mathbb{Z}_2$. Find the irreducible polynomial for $1 + \alpha$ in $\mathbb{Z}_2[x]$.

Concepts

In Exercises 11 through 14, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

11. The vectors in a subset S of a vector space V over a field F *span* V if and only if each $\beta \in V$ can be expressed uniquely as a linear combination of the vectors in S.

12. The vectors in a subset S of a vector space V over a field F are *linearly independent over* F if and only if the zero vector cannot be expressed as a linear combination of vectors in S.

13. The *dimension over* F of a finite-dimensional vector space V over a field F is the minimum number of vectors required to span V.

14. A *basis* for a vector space V over a field F is a set of vectors in V that span V and are linearly dependent.

15. Mark each of the following true or false.

_____ **a.** The sum of two vectors is a vector.
_____ **b.** The sum of two scalars is a vector.
_____ **c.** The product of two scalars is a scalar.
_____ **d.** The product of a scalar and a vector is a vector.
_____ **e.** Every vector space has a finite basis.
_____ **f.** The vectors in a basis are linearly dependent.
_____ **g.** The 0-vector may be part of a basis.
_____ **h.** If $F \leq E$ and $\alpha \in E$ is algebraic over the field F, then α^2 is algebraic over F.
_____ **i.** If $F \leq E$ and $\alpha \in E$ is algebraic over the field F, then $\alpha + \alpha^2$ is algebraic over F.
_____ **j.** Every vector space has a basis.

The exercises that follow deal with the further study of vector spaces. In many cases, we are asked to define for vector spaces some concept that is analogous to one we have studied for other algebraic structures. These exercises should improve our ability to recognize parallel and related situations in algebra. Any of these exercises may assume knowledge of concepts defined in the preceding exercises.

16. Let V be a vector space over a field F.

a. Define a *subspace of the vector space* V *over* F.

b. Prove that an intersection of subspaces of V is again a subspace of V over F.

17. Let V be a vector space over a field F, and let $S = \{\alpha_i \mid i \in I\}$ be a nonempty collection of vectors in V.

a. Using Exercise 16(b), define the *subspace of* V *generated by* S.

b. Prove that the vectors in the subspace of V generated by S are precisely the (finite) linear combinations of vectors in S. (Compare with Theorem 7.6.)

18. Let $V_1, \cdots, V_n$ be vector spaces over the same field F. Define the *direct sum* $V_1 \oplus \cdots \oplus V_n$ *of the vectors spaces* V_i for $i = 1, \cdots, n$, and show that the direct sum is again a vector space over F.

19. Generalize Example 30.2 to obtain the vector space F^n of ordered n-tuples of elements of F over the field F, for any field F. What is a basis for F^n?

20. Define an *isomorphism* of a vector space V over a field F with a vector space V' over the same field F.

Theory

21. Prove that if V is a finite-dimensional vector space over a field F, then a subset $\{\beta_1, \beta_2, \cdots, \beta_n\}$ of V is a basis for V over F if and only if every vector in V can be expressed *uniquely* as a linear combination of the β_i.

22. Let F be any field. Consider the "system of m simultaneous linear equations in n unknowns"

$$\begin{aligned} a_{11}X_1 + a_{12}X_2 + \cdots + a_{1n}X_n &= b_1, \\ a_{21}X_1 + a_{22}X_2 + \cdots + a_{2n}X_n &= b_2, \\ &\vdots \\ a_{m1}X_1 + a_{m2}X_2 + \cdots + a_{mn}X_n &= b_m, \end{aligned}$$

where $a_{ij}, b_i \in F$.

a. Show that the "system has a solution," that is, there exist $X_1, \cdots, X_n \in F$ that satisfy all m equations, if and only if the vector $\beta = (b_1, \cdots, b_m)$ of F^m is a linear combination of the vectors $\alpha_j = (a_{1j}, \cdots, a_{mj})$. (This result is straightforward to prove, being practically the definition of a solution, but should really be regarded as the *fundamental existence theorem for a simultaneous solution of a system of linear equations*.)

b. From part (a), show that if $n = m$ and $\{\alpha_j \mid j = 1, \cdots, n\}$ is a basis for F^n, then the system always has a unique solution.

23. Prove that every finite-dimensional vector space V of dimension n over a field F is isomorphic to the vector space F^n of Exercise 19.

24. Let V and V' be vector spaces over the same field F. A function $\phi : V \to V'$ is a **linear transformation of** V **into** V' if the following conditions are satisfied for all $\alpha, \beta \in V$ and $a \in F$:

$$\phi(\alpha + \beta) = \phi(\alpha) + \phi(\beta).$$
$$\phi(a\alpha) = a(\phi(\alpha)).$$

a. If $\{\beta_i \mid i \in I\}$ is a basis for V over F, show that a linear transformation $\phi : V \to V'$ is completely determined by the vectors $\phi(\beta_i) \in V'$.

b. Let $\{\beta_i \mid i \in I\}$ be a basis for V, and let $\{\beta_i' \mid i \in I\}$ be any set of vectors, not necessarily distinct, of V'. Show that there exists exactly one linear transformation $\phi : V \to V'$ such that $\phi(\beta_i) = \beta_i'$.

25. Let V and V' be vector spaces over the same field F, and let $\phi : V \to V'$ be a linear transformation.

a. To what concept that we have studied for the algebraic structures of groups and rings does the concept of a *linear transformation* correspond?

b. Define the *kernel* (or *nullspace*) of ϕ, and show that it is a subspace of V.

c. Describe when ϕ is an isomorphism of V with V'.

26. Let V be a vector space over a field F, and let S be a subspace of V. Define the *quotient space* V/S, and show that it is a vector space over F.

27. Let V and V' be vector spaces over the same field F, and let V be finite dimensional over F. Let $\dim(V)$ be the dimension of the vector space V over F. Let $\phi : V \to V'$ be a linear transformation.

a. Show that $\phi[V]$ is a subspace of V'.

b. Show that $\dim(\phi[V]) = \dim(V) - \dim(\text{Ker}(\phi))$. [*Hint:* Choose a convenient basis for V, using Theorem 30.19. For example, enlarge a basis for Ker (ϕ) to a basis for V.]

SECTION 31 ALGEBRAIC EXTENSIONS

Finite Extensions

In Theorem 30.23 we saw that if E is an extension field of a field F and $\alpha \in E$ is algebraic over F, then every element of $F(\alpha)$ is algebraic over F. In studying zeros of polynomials in $F[x]$, we shall be interested almost exclusively in extensions of F containing only elements algebraic over F.

31.1 Definition An extension field E of a field F is an **algebraic extension of** F if every element in E is algebraic over F. ■

31.2 Definition If an extension field E of a field F is of finite dimension n as a vector space over F, then E is a **finite extension of degree n over** F. We shall let $[E : F]$ be the degree n of E over F. ■

To say that a field E is a finite extension of a field F does *not* mean that E is finite field. It just asserts that E is a finite-dimensional vector space over F, that is, that $[E : F]$ is finite.

We shall often use the fact that if E is a finite extension of F, then, $[E : F] = 1$ if and only if $E = F$. We need only observe that by Theorem 30.19, $\{1\}$ can always be enlarged to a basis for E over F. Thus $[E : F] = 1$ if and only if $E = F(1) = F$.

Let us repeat the argument of Theorem 30.23 to show that a finite extension E of a field F must be an algebraic extension of F.

31.3 Theorem A finite extension field E of a field F is an algebraic extension of F.

Proof We must show that for $\alpha \in E$, α is algebraic over F. By Theorem 30.19 if $[E : F] = n$, then

$$1, \alpha, \cdots, \alpha^n$$

cannot be linearly independent elements, so there exist $a_i \in F$ such that

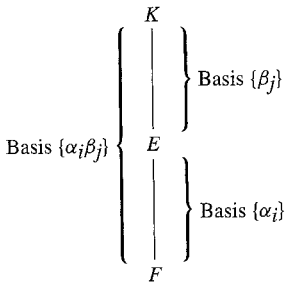
$$a_n\alpha^n + \cdots + a_1\alpha + a_0 = 0,$$

and not all $a_i = 0$. Then $f(x) = a_nx^n + \cdots + a_1x + a_0$ is a nonzero polynomial in $F[x]$. and $f(\alpha) = 0$. Therefore, α is algebraic over F. ◆

We cannot overemphasize the importance of our next theorem. It plays a role in field theory analogous to the role of the theorem of Lagrange in group theory. While its proof follows easily from our brief work with vector spaces, it is a tool of incredible power. An elegant application of it in the section that follows shows the impossibility of performing certain geometric constructions with a straightedge and a compass. *Never underestimate a theorem that counts something.*

31.4 Theorem If E is a finite extension field of a field F, and K is a finite extension field of E, then K is a finite extension of F, and

$$[K : F] = [K : E][E : F].$$



31.5 Figure

Proof Let $\{\alpha_i \mid i = 1, \cdots, n\}$ be a basis for E as a vector space over F, and let the set $\{\beta_j \mid j = 1, \cdots, m\}$ be a basis for K as a vector space over E. The theorem will be proved if we can show that the mn elements $\alpha_i\beta_j$ form a basis for K, viewed as a vector space over F. (See Fig. 31.5.)

Let γ be any element of K. Since the β_j form a basis for K over E, we have

$$\gamma = \sum_{j=1}^{m} b_j\beta_j$$

for $b_j \in E$. Since the α_i form a basis for E over F, we have

$$b_j = \sum_{i=1}^{n} a_{ij}\alpha_i$$

for $a_{ij} \in F$. Then

$$\gamma = \sum_{j=1}^{m}\left(\sum_{i=1}^{n} a_{ij}\alpha_i\right)\beta_j = \sum_{i,j} a_{ij}(\alpha_i\beta_j),$$

so the mn vectors $\alpha_i\beta_j$ span K over F.

It remains for us to show that the mn elements $\alpha_i\beta_j$ are independent over F. Suppose that $\Sigma_{i,j}c_{ij}(\alpha_i\beta_j) = 0$, with $c_{ij} \in F$. Then

$$\sum_{j=1}^{m}\left(\sum_{i=1}^{n} c_{ij}\alpha_i\right)\beta_j = 0,$$

and $(\Sigma_{i=1}^{n} c_{ij}\alpha_i) \in E$. Since the elements β_j are independent over E, we must have

$$\sum_{i=1}^{n} c_{ij}\alpha_i = 0$$

for all j. But now the α_i are independent over F, so $\sum_{i=1}^{n} c_{ij}\alpha_i = 0$ implies that $c_{ij} = 0$ for all i and j. Thus the $\alpha_i\beta_j$ not only span K over F but also are independent over F. Thus they form a basis for K over F. ◆

Note that we proved this theorem by actually exhibiting a basis. It is worth remembering that if $\{\alpha_i \mid i = 1, \cdots, n\}$ is a basis for E over F and $\{\beta_j \mid j = 1, \cdots, m\}$ is a basis for K over E, for fields $F \leq E \leq K$, then the set $\{\alpha_i\beta_j\}$ of mn products is a basis for K over F. Figure 31.5 gives a diagram for this situation. We shall illustrate this further in a moment.

31.6 Corollary If F_i is a field for $i = 1, \cdots, r$ and F_{i+1} is a finite extension of F_i, then F_r is a finite extension of F_1, and

$$[F_r : F_1] = [F_r : F_{r-1}][F_{r-1} : F_{r-2}] \cdots [F_2 : F_1].$$

Proof The proof is a straightforward extension of Theorem 31.4 by induction. ◆

31.7 Corollary If E is an extension field of F, $\alpha \in E$ is algebraic over F, and $\beta \in F(\alpha)$, then $\deg(\beta, F)$ divides $\deg(\alpha, F)$.

Proof By Theorem 30.23, $\deg(\alpha, F) = [F(\alpha) : F]$ and $\deg(\beta, F) = [F(\beta) : F]$. We have $F \leq F(\beta) \leq F(\alpha)$, so by Theorem 31.4 $[F(\beta) : F]$ divides $[F(\alpha) : F]$. ◆

The following example illustrates a type of argument one often makes using Theorem 31.4 or its corollaries.

31.8 Example By Corollary 31.7, there is no element of $\mathbb{Q}(\sqrt{2})$ that is a zero of $x^3 - 2$. Note that $\deg(\sqrt{2}, \mathbb{Q}) = 2$, while a zero of $x^3 - 2$ is of degree 3 over $\mathbb{Q}$, but 3 does not divide 2. ▲

Let E be an extension field of a field F, and let α_1, α_2 be elements of E, not necessarily algebraic over F. By definition, $F(\alpha_1)$ is the smallest extension field of F in E that contains α_1. Similarly, $(F(\alpha_1))(\alpha_2)$ can be characterized as the smallest extension field of F in E containing both α_1 and α_2. We could equally have started with α_2, so $(F(\alpha_1))(\alpha_2) = (F(\alpha_2))(\alpha_1)$. We denote this field by $F(\alpha_1, \alpha_2)$. Similarly, for $\alpha_i \in E$, $F(\alpha_1, \cdots \alpha_n)$ is the smallest extension field of F in E containing all the α_i for $i = 1, \cdots, n$. We obtain the field $F(\alpha_1, \cdots, \alpha_n)$ from the field F by **adjoining to F the elements** α_i in E. Exercise 49 of Section 18 shows that, analogous to an intersection of subgroups of a group, an intersection of subfields of a field E is again a subfield of E. Thus $F(\alpha_1, \cdots, \alpha_n)$ can be characterized as the intersection of all subfields of E containing F and all the α_i for $i = 1, \cdots, n$.

31.9 Example Consider $\mathbb{Q}(\sqrt{2})$. Theorem 30.23 shows that $\{1, \sqrt{2}\}$ is a basis for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$. Using the technique demonstrated in Example 29.10, we can easily discover that $\sqrt{2} + \sqrt{3}$ is a zero of $x^4 - 10x^2 + 1$. By the method demonstrated in Example 23.14, we can show that this polynomial is irreducible in $\mathbb{Q}[x]$. Thus $\text{irr}(\sqrt{2} + \sqrt{3}, \mathbb{Q}) = x^4 - 10x^2 + 1$, so $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$. Thus $(\sqrt{2} + \sqrt{3}) \notin \mathbb{Q}(\sqrt{2})$, so $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$. Consequently, $\{1, \sqrt{3}\}$ is a basis for $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = (\mathbb{Q}(\sqrt{2}))(\sqrt{3})$ over $\mathbb{Q}(\sqrt{2})$. The proof of Theorem 31.4 (see the comment following the theorem) then shows that $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ is a basis for $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$. ▲

31.10 Example Let $2^{1/3}$ be the real cube root of 2 and $2^{1/2}$ be the positive square root of 2. Then $2^{1/2} \notin \mathbb{Q}(2^{1/3})$ because $\deg(2^{1/2}, \mathbb{Q}) = 2$ and 2 is not a divisor of $3 = \deg(2^{1/3}, \mathbb{Q})$. Thus $[\mathbb{Q}(2^{1/3}, 2^{1/2}) : \mathbb{Q}(2^{1/3})] = 2$. Hence $\{1, 2^{1/3}, 2^{2/3}\}$ is a basis for $\mathbb{Q}(2^{1/3})$ over $\mathbb{Q}$ and $\{1, 2^{1/2}\}$ is a basis for $\mathbb{Q}(2^{1/3}, 2^{1/2})$ over $\mathbb{Q}(2^{1/3})$. Furthermore, by Theorem 31.4 (see the comment following the theorem),

$$\{1, 2^{1/2}, 2^{1/3}, 2^{5/6}, 2^{2/3}, 2^{7/6}\}$$

is a basis for $\mathbb{Q}(2^{1/2}, 2^{1/3})$ over $\mathbb{Q}$. Because $2^{7/6} = 2(2^{1/6})$, we have $2^{1/6} \in \mathbb{Q}(2^{1/2}, 2^{1/3})$. Now $2^{1/6}$ is a zero of $x^6 - 2$, which is irreducible over $\mathbb{Q}$, by Eisenstein's criterion, with

$p = 2$. Thus

$$\mathbb{Q} \leq \mathbb{Q}(2^{1/6}) \leq \mathbb{Q}(2^{1/2}, 2^{1/3})$$

and by Theorem 31.4

$$\begin{aligned} 6 &= [\mathbb{Q}(2^{1/2}, 2^{1/3}) : \mathbb{Q}] = [\mathbb{Q}(2^{1/2}, 2^{1/3}) : \mathbb{Q}(2^{1/6})][\mathbb{Q}(2^{1/6}) : \mathbb{Q}] \\ &= [\mathbb{Q}(2^{1/2}, 2^{1/3}) : \mathbb{Q}(2^{1/6})](6). \end{aligned}$$

Therefore, we must have

$$[\mathbb{Q}(2^{1/2}, 2^{1/3}) : \mathbb{Q}(2^{1/6})] = 1,$$

so $\mathbb{Q}(2^{1/2}, 2^{1/3}) = \mathbb{Q}(2^{1/6})$, by the comment preceding Theorem 31.3. ▲

Example 31.10 shows that it is possible for an extension $F(\alpha_1, \cdots, \alpha_n)$ of a field F to be actually a simple extension, even though $n > 1$.

Let us characterize extensions of F of the form $F(\alpha_1, \cdots, \alpha_n)$ in the case that all the α_i are algebraic over F.

31.11 Theorem Let E be an algebraic extension of a field F. Then there exist a finite number of elements $\alpha_1, \cdots, \alpha_n$ in E such that $E = F(\alpha_1, \cdots, \alpha_n)$ if and only if E is a finite-dimensional vector space over F, that is, if and only if E is a finite extension of F.

Proof Suppose that $E = F(\alpha_1, \cdots, \alpha_n)$. Since E is an algebraic extension of F, each α_i, is algebraic over F, so each α_i is algebraic over every extension field of F in E. Thus $F(\alpha_1)$ is algebraic over F, and in general, $F(\alpha_1, \cdots, \alpha_j)$ is algebraic over $F(\alpha_i, \cdots, \alpha_{j-1})$ for $j = 2, \cdots, n$. Corollary 31.6 applied to the sequence of finite extensions

$$F, F(\alpha_1), F(\alpha_1, \alpha_2), \cdots, F(\alpha_1, \cdots, \alpha_n) = E$$

then shows that E is a finite extension of F.

Conversely, suppose that E is a finite algebraic extension of F. If $[E : F] = 1$, then $E = F(1) = F$, and we are done. If $E \neq F$, let $\alpha_1 \in E$, where $\alpha_1 \notin F$. Then $[F(\alpha_1) : F] > 1$. If $F(\alpha_1) = E$, we are done; if not, let $\alpha_2 \in E$, where $\alpha_2 \notin F(\alpha_1)$. Continuing this process, we see from Theorem 31.4 that since $[E : F]$ is finite, we must arrive at α_n such that

$$F(\alpha_1, \cdots, \alpha_n) = E.$$ ◆

Algebraically Closed Fields and Algebraic Closures

We have not yet observed that if E is an extension of a field F and $\alpha, \beta \in E$ are algebraic over F, then so are $\alpha + \beta$, $\alpha\beta$, $\alpha - \beta$, and α/β, if $\beta \neq 0$. This follows from Theorem 31.3 and is also included in the following theorem.

31.12 Theorem Let E be an extension field of F. Then

$$\bar{F}_E = \{\alpha \in E \mid \alpha \text{ is algebraic over } F\}$$

is a subfield of E, the **algebraic closure of F in E**.

Proof Let $\alpha, \beta \in \bar{F}_E$. Then Theorem 31.11 shows that $F(\alpha, \beta)$ is a finite extension of F, and by Theorem 31.3 every element of $F(\alpha, \beta)$ is algebraic over F, that is, $F(\alpha, \beta) \subseteq \bar{F}_E$. Thus

$\bar{F}_E$ contains $\alpha + \beta, \alpha\beta, \alpha - \beta$, and also contains α/β for $\beta \neq 0$, so $\bar{F}_E$ is a subfield of E. ◆

31.13 Corollary The set of all algebraic numbers forms a field.

Proof Proof of this corollary is immediate from Theorem 31.12, because the set of all algebraic numbers is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. ◆

It is well known that the complex numbers have the property that every nonconstant polynomial in $\mathbb{C}[x]$ has a zero in $\mathbb{C}$. This is known as the *Fundamental Theorem of Algebra*. An analytic proof of this theorem is given in Theorem 31.18. We now give a definition generalizing this important concept to other fields.

31.14 Definition A field F is **algebraically closed** if every nonconstant polynomial in $F[x]$ has a zero in F. ■

Note that a field F can be the algebraic closure of F in an extension field E without F being algebraically closed. For example, $\mathbb{Q}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{Q}(x)$, but $\mathbb{Q}$ is not algebraically closed because $x^2 + 1$ has no zero in $\mathbb{Q}$.

The next theorem shows that the concept of a field being algebraically closed can also be defined in terms of factorization of polynomials over the field.

31.15 Theorem A field F is algebraically closed if and only if every nonconstant polynomial in $F[x]$ factors in $F[x]$ into linear factors.

Proof Let F be algebraically closed, and let $f(x)$ be a nonconstant polynomial in $F[x]$ Then $f(x)$ has a zero $a \in F$. By Corollary 23.3, $x - a$ is a factor of $f(x)$, so $f(x) = (x - a)g(x)$. Then if $g(x)$ is nonconstant, it has a zero $b \in F$, and we have $f(x) = (x - a)(x - b)h(x)$. Continuing, we get a factorization of $f(x)$ in $F[x]$ into linear factors.

Conversely, suppose that every nonconstant polynomial of $F[x]$ has a factorization into linear factors. If $ax - b$ is a linear factor of $f(x)$, then b/a is a zero of $f(x)$. Thus F is algebraically closed. ◆

31.16 Corollary An algebraically closed field F has no proper algebraic extensions, that is, no algebraic extensions E with $F < E$.

Proof Let E be an algebraic extension of F, so $F \leq E$. Then if $\alpha \in E$, we have $\text{irr}(\alpha, F) = x - \alpha$, by Theorem 31.15, since F is algebraically closed. Thus $\alpha \in F$, and we must have $F = E$. ◆

In a moment we shall show that just as there exists an algebraically closed extension $\mathbb{C}$ of the real numbers $\mathbb{R}$, for any field F there exists similarly an algebraic extension $\bar{F}$ of F, with the property that $\bar{F}$ is algebraically closed. Naively, to find $\bar{F}$ we proceed as follows. If a polynomial $f(x)$ in $F[x]$ has a no zero in F, then adjoin a zero α of such an $f(x)$ to F, thus obtaining the field $F(\alpha)$. *Theorem 29.3, Kronecker's theorem, is strongly used here, of course.* If $F(\alpha)$ is still not algebraically closed, then continue the process further. The trouble is that, contrary to the situation for the algebraic closure $\mathbb{C}$ of $\mathbb{R}$, we may have to do this a (possibly large) infinite number of times. It can be shown (see Exercises 33 and 36) that $\overline{\mathbb{Q}}$ is isomorphic to the field of all algebraic numbers, and that

we cannot obtain $\overline{\mathbb{Q}}$ from $\mathbb{Q}$ by adjoining a finite number of algebraic numbers. We shall have to first discuss some set-theoretic machinery, *Zorn's lemma*, in order to be able to handle such a situation. This machinery is a bit complex, so we are putting the proof under a separate heading. The existence theorem for $\bar{F}$ is very important, and we state it here so that we will know this fact, even if we do not study the proof.

31.17 Theorem Every field F has an **algebraic closure**, that is, an algebraic extension $\bar{F}$ that is algebraically closed.

It is well known that $\mathbb{C}$ is an algebraically closed field. We recall an analytic proof for the student who has had a course in functions of a complex variable. There are algebraic proofs, but they are much longer.

31.18 Theorem **(Fundamental Theorem of Algebra)** The field $\mathbb{C}$ of complex numbers is an algebraically closed field.

Proof Let the polynomial $f(z) \in \mathbb{C}[z]$ have no zero in $\mathbb{C}$. Then $1/f(z)$ gives an entire function; that is, $1/f$ is analytic everywhere. Also if $f \notin \mathbb{C}$, $\lim_{|z|\to\infty} |f(z)| = \infty$, so $\lim_{|z|\to\infty} |1/f(z)| = 0$. Thus $1/f$ must be bounded in the plane. Hence by Liouville's theorem of complex function theory, $1/f$ is constant, and thus f is constant. Therefore, a nonconstant polynomial in $\mathbb{C}[z]$ must have a zero in $\mathbb{C}$, so $\mathbb{C}$ is algebraically closed. ◆

Proof of the Existence of an Algebraic Closure

We shall prove that every field has an algebraic extension that is algebraically closed. Mathematics students should have the opportunity to see some proof involving the *Axiom of Choice* by the time they finish college. This is a natural place for such a proof. We shall use an equivalent form, *Zorn's lemma*, of the Axiom of Choice. To state Zorn's lemma, we have to give a set-theoretic definition.

31.19 Definition A **partial ordering of a set** S is given by a relation $\leq$ defined for certain ordered pairs of elements of S such that the following conditions are satisfied:

1. $a \leq a$ for all $a \in S$ **(reflexive law).**
2. If $a \leq b$ and $b \leq a$, then $a = b$ **(antisymmetric law).**
3. If $a \leq b$ and $b \leq c$, then $a \leq c$ **(transitive law).** ■

In a *partially* ordered set, not every two elements need be **comparable;** that is, for $a, b \in S$, we need not have either $a \leq b$ or $b \leq a$. As usual, $a < b$ denotes $a \leq b$ but $a \neq b$.

A subset T of a partially ordered set S is a **chain** if every two elements a and b in T are comparable, that is, either $a \leq b$ or $b \leq a$ (or both). An element $u \in S$ is an **upper bound for a subset** A of partially ordered set S if $a \leq u$ for all $a \in A$. Finally, an element m of a partially ordered set S is **maximal** if there is no $s \in S$ such that $m < s$.

31.20 Example The collection of all subsets of a set forms a partially ordered set under the relation $\leq$ given by $\subseteq$. For example, if the whole set is $\mathbb{R}$, we have $\mathbb{Z} \subseteq \mathbb{Q}$. Note, however, that for $\mathbb{Z}$ and $\mathbb{Q}^+$, neither $\mathbb{Z} \subseteq \mathbb{Q}^+$ nor $\mathbb{Q}^+ \subseteq \mathbb{Z}$. ▲

31.21 Zorn's Lemma If S is a partially ordered set such that every chain in S has an upper bound in S, then S has at least one maximal element.

There is no question of *proving* Zorn's lemma. The lemma is equivalent to the Axiom of Choice. Thus we are really taking Zorn's lemma here as an *axiom* for our set theory. Refer to the literature for a statement of the Axiom of Choice and a proof of its equivalence to Zorn's lemma. (See Edgerton [47].)

Zorn's lemma is often useful when we want to show the existence of a largest or maximal structure of some kind. If a field F has an algebraic extension $\bar{F}$ that is algebraically closed, then $\bar{F}$ will certainly be a maximal algebraic extension of F, for since $\bar{F}$ is algebraically closed, it can have no proper algebraic extensions.

The idea of our proof of Theorem 31.17 is very simple. Given a field F, we shall first describe a class of algebraic extensions of F that is so large that it must contain (up to isomorphism) any conceivable algebraic extension of F. We then define a partial ordering, the ordinary subfield ordering, on this class, and show that the hypotheses of Zorn's lemma are satisfied. By Zorn's lemma, there will exist a maximal algebraic extension $\bar{F}$ of F in this class. We shall then argue that, as a maximal element, this extension $\bar{F}$ can have no proper algebraic extensions, so it must be algebraically closed.

Our proof differs a bit from the one found in many texts. We like it because it uses no algebra other than that derived from Theorems 29.3 and 31.4. Thus it throws into sharp relief the tremendous strength of both Kronecker's theorem and Zorn's lemma. The proof looks long, but only because we are writing out every little step. To the professional mathematician, the construction of the proof from the information in the preceding paragraph is a routine matter. This proof was suggested to the author during his graduate student days by a fellow graduate student, Norman Shapiro, who also had a strong preference for it.

■ Historical Note

The Axiom of Choice, although used implicitly in the 1870s and 1880s, was first stated explicitly by Ernst Zermelo in 1904 in connection with his proof of the well-ordering theorem, the result that for any set A, there exists an order–relation $<$ such that every nonempty subset B of A contains a least element with respect to $<$. Zermelo's Axiom of Choice asserted that, given any set M and the set S of all subsets of M, there always exists a "choice" function, a function $f : S \to M$ such that $f(M') \in M'$ for every M' in S. Zermelo noted, in fact, that "this logical principal cannot... be reduced to a still simpler one, but it is applied without hesitation everywhere in mathematical deduction." A few years later he included this axiom in his collection of axioms for set theory, a collection which was slightly modified in 1930 into what is now called Zermelo–Fraenkel set theory, the axiom system generally used today as a basis of that theory.

Zorn's lemma was introduced by Max Zorn (1906–1993) in 1935. Although he realized that it was equivalent to the well-ordering theorem (itself equivalent to the Axiom of Choice), he claimed that his lemma was more natural to use in algebra because the well-ordering theorem was somehow a "transcendental" principal. Other mathematicians soon agreed with his reasoning. The lemma appeared in 1939 in the first volume of Nicolas Bourbaki's *Eléments de Mathématique: Les Structures Fondamentales de l'Analyse*. It was used consistently in that work and quickly became an essential part of the mathematician's toolbox.

We are now ready to carry out our proof of Theorem 31.17, which we restate here.

31.22 Restated Theorem 31.17 Every field F has an algebraic closure $\bar{F}$.

Proof It can be shown in set theory that given any set, there exists a set with *strictly more* elements. Suppose we form a set

$$A = \{\omega_{f,i} \mid f \in F[x]; i = 0, \cdots, (\text{degree } f)\}$$

that has an element for every possible zero of any $f(x) \in F[x]$. Let Ω be a set with strictly more elements than A. Replacing Ω by $\Omega \cup F$ if necessary, we can assume $F \subset \Omega$. Consider all possible fields that are algebraic extension of F and that, as sets, consist of elements of Ω. One such algebraic extension is F itself. If E is any extension field of F, and if $\gamma \in E$ is a zero $f(x) \in F[x]$ for $\gamma \notin F$ and $\deg(\gamma, F) = n$, then renaming γ by ω for $\omega \in \Omega$ and $\omega \notin F$, and renaming elements $a_0 + a_1\gamma + \cdots + a_{n-1}\gamma^{n-1}$ of $F(\gamma)$ by distinct elements of Ω as the a_i range over F, we can consider our renamed $F(\gamma)$ to be an algebraic extension field $F(\omega)$ of F, with $F(\omega) \subset \Omega$ and $f(\omega) = 0$. The set Ω has enough elements to form $F(\omega)$, since Ω has more than enough elements to provide n different zeros for each element of each degree n in any subset of $F[x]$.

All algebraic extension fields E_j of F, with $E_j \subseteq \Omega$, form a set

$$S = \{E_j \mid j \in J\}$$

that is partially ordered under our usual subfield inclusion $\leq$. One element of S is F itself. The preceding paragraphs shows that if F is far away from being algebraically closed, there will be many fields E_j in S.

Let $T = \{E_{j_k}\}$ be a chain in S, and let $W = \cup_k E_{j_k}$. We now make W into a field. Let $\alpha, \beta \in W$. Then there exist $E_{j_1}, E_{j_2} \in S$, with $\alpha \in E_{j_1}$ and $\beta \in E_{j_2}$. Since T is a chain, one of the fields E_{j_1} and E_{j_2} is a subfield of the other, say $E_{j_1} \leq E_{j_2}$. Then $\alpha, \beta \in E_{j_2}$, and we use the field operations of E_{j_2} to *define* the sum of α and β in W as $(\alpha + \beta) \in E_{j_2}$ and, likewise, the product as $(\alpha\beta) \in E_{j_2}$. These operations are well defined in W; they are independent of our choice of E_{j_2}, since if $\alpha, \beta \in E_{j_3}$ also, for E_{j_3} in T, then one of the fields E_{j_2} and E_{j_3} is a subfield of the other, since T is a chain. Thus we have operations of addition and multiplication defined on W.

All the field axioms for W under these operations now follow from the fact that these operations were defined in terms of addition and multiplication in fields. Thus, for example, $1 \in F$ serves as multiplicative identity in W, since for $\alpha \in W$, if $1, \alpha \in E_{j_1}$, then we have $1\alpha = \alpha$ in E_{j_1}, so $1\alpha = \alpha$ in W, by definition of multiplication in W. As further illustration, to check the distributive laws, let $\alpha, \beta, \gamma \in W$. Since T is a chain, we can find one field in T containing all three elements α, β, and γ, and in this field the distributive laws for α, β, and γ hold. Thus they hold in W. Therefore, we can view W as a field, and by construction, $E_{j_k} \leq W$ for every $E_{j_k} \in T$.

If we can show that W is algebraic over F, then $W \in S$ will be an upper bound for T. But if $\alpha \in W$, then $\alpha \in E_{j_1}$ for some E_{j_1} in T, so α is algebraic over F. Hence W is an algebraic extension of F and is an upper bound for T.

The hypotheses of Zorn's lemma are thus fulfilled, so there is a maximal element $\bar{F}$ of S. We claim that $\bar{F}$ is algebraically closed. Let $f(x) \in \bar{F}[x]$, where $f(x) \notin \bar{F}$. Suppose that $f(x)$ has no zero in $\bar{F}$. Since Ω has many more elements than $\bar{F}$ has, we

can take $\omega \in \Omega$, where $\omega \notin \bar{F}$, and form a field $\bar{F}(\omega) \subseteq \Omega$, with ω a zero of $f(x)$, as we saw in the first paragraph of this proof. Let β be in $\bar{F}(\omega)$. Then by Theorem 30.23, β is a zero of a polynomial

$$g(x) = \alpha_0 + \alpha_1 x + \cdots + \alpha_n x^n$$

in $\bar{F}[x]$, with $\alpha_i \in \bar{F}$, and hence α_i algebraic over F. Then by Theorem 31.11 the field $F(\alpha_0, \cdots, \alpha_n)$ is a finite extension of F, and since β is algebraic over $F(\alpha_0, \cdots, \alpha_n)$, we also see that $F(\alpha_0, \cdots, \alpha_n, \beta)$ is a finite extension over $F(\alpha_0, \cdots, \alpha_n)$. Theorem 31.4 then shows that $F(\alpha_0, \cdots, \alpha_n, \beta)$ is a finite extension of F, so by Theorem 31.3, β is algebraic over F. Hence $\bar{F}(\omega) \in S$ and $\bar{F} < \bar{F}(\omega)$, which contradicts the choice of $\bar{F}$ as maximal in S. Thus $f(x)$ must have had a zero in $\bar{F}$, so $\bar{F}$ is algebraically closed. ◆

The mechanics of the preceding proof are routine to the professional mathematician. Since it may be the first proof that we have ever seen using Zorn's lemma, we wrote the proof out in detail.

EXERCISES 31

Computations

In Exercises 1 through 13, find the degree and a basis for the given field extension. Be prepared to justify your answers.

1. $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$

2. $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$

3. $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$

4. $\mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$ over $\mathbb{Q}$

5. $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$ over $\mathbb{Q}$

6. $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ over $\mathbb{Q}$

7. $\mathbb{Q}(\sqrt{2}\sqrt{3})$ over $\mathbb{Q}$

8. $\mathbb{Q}(\sqrt{2}, \sqrt[3]{5})$ over $\mathbb{Q}$

9. $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{6}, \sqrt[3]{24})$ over $\mathbb{Q}$

10. $\mathbb{Q}(\sqrt{2}, \sqrt{6})$ over $\mathbb{Q}(\sqrt{3})$

11. $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ over $\mathbb{Q}(\sqrt{3})$

12. $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}(\sqrt{2} + \sqrt{3})$

13. $\mathbb{Q}(\sqrt{2}, \sqrt{6} + \sqrt{10})$ over $\mathbb{Q}(\sqrt{3} + \sqrt{5})$

Concepts

In Exercises 14 through 17, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

14. An *algebraic extension* of a field F is a field $F(\alpha_1, \alpha_2, \cdots, \alpha_n)$ where each α_i is a zero of some polynomial in $F[x]$.

15. A *finite extension field* of a field F is one that can be obtained by adjoining a finite number of elements to F.

16. The *algebraic closure $\bar{F}_E$ of a field F in an extension field E of F* is the field consisting of all elements of E that are algebraic over F.

17. A field F is *algebraically closed* if and only if every polynomial has a zero in F.

18. Show by an example that for a proper extension field E of a field F, the algebraic closure of F in E need not be algebraically closed.

19. Mark each of the following true or false.

_____ **a.** If a field E is a finite extension of a field F, then E is a finite field.
_____ **b.** Every finite extension of a field is an algebraic extension.
_____ **c.** Every algebraic extension of a field is a finite extension.
_____ **d.** The top field of a finite tower of finite extensions of fields is a finite extension of the bottom field.
_____ **e.** $\mathbb{Q}$ is its own algebraic closure in $\mathbb{R}$, that is $\mathbb{Q}$ is **algebraically closed in** $\mathbb{R}$.
_____ **f.** $\mathbb{C}$ is algebraically closed in $\mathbb{C}(x)$, where x is an indeterminate.
_____ **g.** $\mathbb{C}(x)$ is algebraically closed, where x is an indeterminate.
_____ **h.** The field $\mathbb{C}(x)$ has no algebraic closure, since $\mathbb{C}$ already contains all algebraic numbers.
_____ **i.** An algebraically closed field must be of characteristic 0.
_____ **j.** If E is an algebraically closed extension field of F, then E is an algebraic extension of F.

Proof Synopsis

20. Give a one-sentence synopsis of the proof of Theorem 31.3.

21. Give a one- or two-sentence synopsis of the proof of Theorem 31.4.

Theory

22. Let $(a + bi) \in \mathbb{C}$ where $a, b \in \mathbb{R}$ and $b \neq 0$. Show that $\mathbb{C} = \mathbb{R}(a + bi)$.

23. Show that if E is a finite extension of a field F and $[E : F]$ is a prime number, then E is a simple extension of F and, indeed, $E = F(\alpha)$ for every $\alpha \in E$ not in F.

24. Prove that $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$.

25. What degree field extensions can we obtain by successively adjoining to a field F a square root of an element of F not a square in F, then square root of some nonsquare in this new field, and so on? Argue from this that a zero of $x^{14} - 3x^2 + 12$ over $\mathbb{Q}$ can never be expressed as a rational function of square roots of rational functions of square roots, and so on, of elements of $\mathbb{Q}$.

26. Let E be a finite extension field of F. Let D be an integral domain such that $F \subseteq D \subseteq E$. Show that D is a field.

27. Prove in detail that $\mathbb{Q}(\sqrt{3} + \sqrt{7}) = \mathbb{Q}(\sqrt{3}, \sqrt{7})$.

28. Generalizing Exercise 27, show that if $\sqrt{a} + \sqrt{b} \neq 0$, then $\mathbb{Q}(\sqrt{a} + \sqrt{b}) = \mathbb{Q}(\sqrt{a}, \sqrt{b})$ for all a and b in $\mathbb{Q}$. [*Hint:* Compute $(a - b)/(\sqrt{a} + \sqrt{b})$.]

29. Let E be a finite extension of a field F, and let $p(x) \in F[x]$ be irreducible over F and have degree that is not a divisor of $[E : F]$. Show that $p(x)$ has no zeros in E.

30. Let E be an extension field of F. Let $\alpha \in E$ be algebraic of odd degree over F. Show that α^2 is algebraic of odd degree over F, and $F(\alpha) = F(\alpha^2)$.

31. Show that if F, E, and K are fields with $F \leq E \leq K$, then K is algebraic over F if and only if E is algebraic over F, and K is algebraic over E. (You must *not* assume the extensions are finite.)

32. Let E be an extension field of a field F. Prove that every $\alpha \in E$ that is not in the algebraic closure $\bar{F}_E$ of F in E is transcendental over $\bar{F}_E$.

33. Let E be an algebraically closed extension field of a field F. Show that the algebraic closure $\bar{F}_E$ of F in E is algebraically closed. (Applying this exercise to $\mathbb{C}$ and $\mathbb{Q}$, we see that the field of all algebraic numbers is an algebraically closed field.)

34. Show that if E is an algebraic extension of a field F and contains all zeros in $\bar{F}$ of every $f(x) \in F[x]$, then E is an algebraically closed field.

35. Show that no finite field of odd characteristic is algebraically closed. (Actually, no finite field of characteristic 2 is algebraically closed either.) [*Hint*: By counting, show that for such a finite field F, some polynomial $x^2 - a$, for some $a \in F$, has no zero in F. See Exercise 32, Section 29.]

36. Prove that, as asserted in the text, the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$ is not a finite extension of $\mathbb{Q}$.

37. Argue that every finite extension field of $\mathbb{R}$ is either $\mathbb{R}$ itself or is isomorphic to $\mathbb{C}$.

38. Use Zorn's lemma to show that every proper ideal of a ring R with unity is contained in some maximal ideal.

SECTION 32 †GEOMETRIC CONSTRUCTIONS

In this section we digress briefly to give an application demonstrating the power of Theorem 31.4. For a more detailed study of geometric constructions, you are referred to Courant and Robbins [44, Chapter III].

We are interested in what types of figures can be constructed with a compass and a straightedge in the sense of classical Euclidean plane geometry. We shall discuss the impossibility of trisecting certain angles and other classical questions.

Constructible Numbers

Let us imagine that we are given only a single line segment that we shall define to be *one unit* in length. A real number α is **constructible** if we can construct a line segment of length $|\alpha|$ in a finite number of steps from this given segment of unit length by using a straightedge and a compass.

The rules of the game are pretty strict. We suppose that we are given just two points at the moment, the endpoints of our unit line segment, let us suppose that they correspond to the points (0, 0) and (1, 0) in the Euclidean plane. We are allowed to draw a line only with our straightedge through two points that we have already located. Thus we can start by using the straightedge and drawing the line through (0, 0) and (1, 0). We are allowed to open our compass only to a distance between points we have already found. Let us open our compass to the distance between (0, 0) and (1, 0). We can then place the point of the compass at (1, 0) and draw a circle of radius 1, which passes through the point (2, 0). Thus we now have located a third point, (2, 0). Continuing in this way, we can locate points (3, 0), (4, 0), (−1, 0), (−2, 0), and so on. Now open the compass the distance from (0, 0) to (0, 2), put the point at (1, 0), and draw a circle of radius 2. Do the same with the point at (−1, 0). We have now found two new points, where these circles intersect, and we can put our straightedge on them to draw what we think of as the y-axis. Then opening our compass to the distance from (0, 0) to (1, 0), we draw a circle with center at (0, 0) and locate the point (0, 1) where the circle intersects the y-axis. Continuing in this fashion, we can locate all points (x, y) with integer coordinates in any rectangle containing the point (0, 0). Without going into more detail, it can be shown that it is possible, among other things, to erect a perpendicular to a given line at a known point

† This chapter is not used in the remainder of the text.

on the line, and find a line passing through a known point and parallel to a given line. Our first result is the following theorem.

32.1 Theorem If α and β are constructible real numbers, then so are $\alpha + \beta$, $\alpha - \beta$, $\alpha\beta$, and α/β, if $\beta \neq 0$.

Proof We are given that α and β are constructible, so there are line segments of lengths $|\alpha|$ and $|\beta|$ available to us. For $\alpha, \beta > 0$, extend a line segment of length α with the straightedge. Start at one end of the original segment of length α, and lay off on the extension the length β with the compass. This constructs a line segment of length $\alpha + \beta$; $\alpha - \beta$ is similiarly constructible (see Fig. 32.2). If α and β are not both positive, an obvious breakdown into cases according to their signs shows that $\alpha + \beta$ and $\alpha - \beta$ are still constructible.

The construction of $\alpha\beta$ is indicated in Fig. 32.3. We shall let $\overline{OA}$ be the line segment from the point O to the point A, and shall let $|\overline{OA}|$ be the length of this line segment. If $\overline{OA}$ is of length $|\alpha|$, construct a line l through O not containing $\overline{OA}$. (Perhaps, if O is at (0, 0) and A is at $(a, 0)$, you use the line through (0, 0) and (4, 2).) Then find the points P and B on l such that $\overline{OP}$ is of length 1 and $\overline{OB}$ is of length $|\beta|$. Draw $\overline{PA}$ and construct l' through B, parallel to $\overline{PA}$ and intersecting $\overline{OA}$ extended at Q. By similar triangles, we have

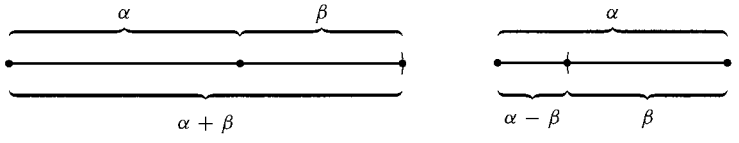
$$\frac{1}{|\alpha|} = \frac{|\beta|}{|\overline{OQ}|},$$

so $\overline{OQ}$ is of length $|\alpha\beta|$.

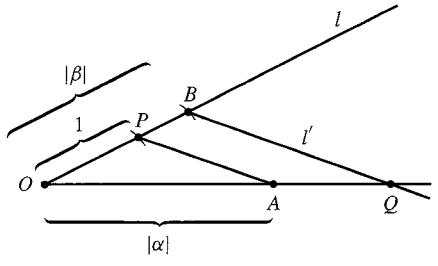
Finally, Fig. 32.4 shows that α/β is constructible if $\beta \neq 0$. Let $\overline{OA}$ be of length $|\alpha|$, and construct l through O not containing OA. Then find B and P on l such that $\overline{OB}$ is of length $|\beta|$ and $\overline{OP}$ is of length 1. Draw BA and construct l' through P, parallel to $\overline{BA}$, and intersecting $\overline{OA}$ at Q. Again by similar triangles, we have

$$\frac{|\overline{OQ}|}{1} = \frac{|\alpha|}{|\beta|},$$

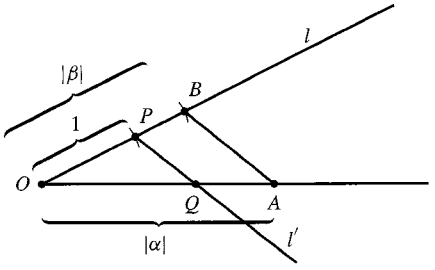
so $\overline{OQ}$ is of length $|\alpha/\beta|$. ◆



32.2 Figure



32.3 Figure



32.4 Figure

32.5 Corollary The set of all constructible real numbers forms a subfield F of the field of real numbers.

Proof Proof of this corollary is immediate from Theorem 32.1. ◆

Thus the field F of all constructible real numbers contains $\mathbb{Q}$, the field of rational numbers, since $\mathbb{Q}$ is the smallest subfield of $\mathbb{R}$.

From now on, we proceed analytically. We can construct any rational number. Regarding our given segment

$$0\text{_____}1$$

of length 1 as the basic unit on an x-axis, we can locate any point (q_1, q_2) in the plane with both coordinates rational. Any further point in the plane that we can locate by using a compass and a straightedge can be found in one of the following three ways:

1. as an intersection of two lines, each of which passes through two known points having rational coordinates,
2. as an intersection of a line that passes through two points having rational coordinates and a circle whose center has rational coordinates and whose radius is rational.
3. as an intersection of two circles whose centers have rational coordinates and whose radii are rational.

Equations of lines and circles of the type discussed in 1, 2, and 3 are of the form

$$ax + by + c = 0$$

and

$$x^2 + y^2 + dx + ey + f = 0,$$

where a, b, c, d, e, and f are all in $\mathbb{Q}$. Since in Case 3 the intersection of two circles with equations

$$x^2 + y^2 + d_1x + e_1y + f_1 = 0$$

and

$$x^2 + y^2 + d_2x + e_2y + f_2 = 0$$

is the same as the intersection of the first circle having equation

$$x^2 + y^2 + d_1x + e_1y + f_1 = 0,$$

and the line (the common chord) having equation

$$(d_1 - d_2)x + (e_1 - e_2)y + f_1 - f_2 = 0,$$

we see that Case 3 can be reduced to Case 2. For Case 1, a simultaneous solution of two linear equations with rational coefficients can only lead to rational values of x and y, giving us no new points. However, finding a simultaneous solution of a linear equation with rational coefficients and a quadratic equation with rational coefficients, as in Case 2, leads, upon substitution, to a quadratic equation. Such an equation, when solved by the quadratic formula, may have solutions involving square roots of numbers that are not squares in $\mathbb{Q}$.

In the preceding argument, nothing was really used involving $\mathbb{Q}$ except field axioms. If H is the smallest field containing those real numbers constructed so far, the argument shows that the "next new number" constructed lies in a field $H(\sqrt{\alpha})$ for some $\alpha \in H$, where $\alpha > 0$. We have proved half of our next theorem.

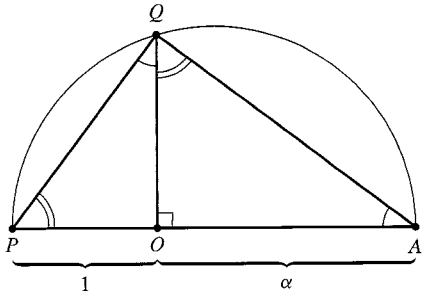
32.6 Theorem The field F of constructible real numbers consists precisely of all real numbers that we can obtain from $\mathbb{Q}$ by taking square roots of positive numbers a finite number of times and applying a finite number of field operations.

Proof We have shown that F can contain no numbers except those we obtain from $\mathbb{Q}$ by taking a finite number of square roots of positive numbers and applying a finite number of field operations. However, if $\alpha > 0$ is constructible, then Fig. 32.7 shows that $\sqrt{\alpha}$ is constructible. Let $\overline{OA}$ have length α, and find P on $\overline{OA}$ extended so that $\overline{OP}$ has length 1. Find the midpoint of $\overline{PA}$ and draw a semicircle with $\overline{PA}$ as diameter. Erect a perpendicular to $\overline{PA}$ at O, intersecting the semicircle at Q. Then the triangles OPQ and OQA are similar, so

$$\frac{|\overline{OQ}|}{|\overline{OA}|} = \frac{|\overline{OP}|}{|\overline{OQ}|},$$

and $|\overline{OQ}|^2 = 1\alpha = \alpha$. Thus $\overline{OQ}$ is of length $\sqrt{\alpha}$. Therefore square roots of constructible numbers are constructible.

Theorem 32.1 showed that field operations are possible by construction. ◆



32.7 Figure

32.8 Corollary If γ is constructible and $\gamma \notin \mathbb{Q}$, then there is a finite sequence of real numbers $\alpha_1, \cdots, \alpha_n = \gamma$ such that $\mathbb{Q}(\alpha_1, \cdots, \alpha_i)$ is an extension of $\mathbb{Q}(\alpha_1, \cdots, \alpha_{i-1})$ of degree 2. In particular, $[\mathbb{Q}(\gamma) : \mathbb{Q}] = 2^r$ for some integer $r \geq 0$.

Proof The existence of the α_i is immediate from Theorem 32.6. Then

$$\begin{aligned} 2^n &= [\mathbb{Q}(\alpha_1, \cdots, \alpha_n) : \mathbb{Q}] \\ &= [\mathbb{Q}(\alpha_1, \cdots, \alpha_n) : \mathbb{Q}(\gamma)][\mathbb{Q}(\gamma) : \mathbb{Q}], \end{aligned}$$

by Theorem 32.4, which completes the proof. ◆

The Impossibility of Certain Constructions

We can now show the impossibility of certain geometric constructions.

32.9 Theorem *Doubling the cube is impossible,* that is, given a side of a cube, it is not always possible to construct with a straightedge and a compass the side of a cube that has double the volume of the original cube.

Proof Let the given cube have a side of length 1, and hence a volume of 1. The cube being sought would have to have a volume of 2, and hence a side of length $\sqrt[3]{2}$. But $\sqrt[3]{2}$ is a zero of irreducible $x^3 - 2$ over $\mathbb{Q}$, so

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3.$$

Corollary 32.8 shows that to double this cube of volume 1, we would need to have $3 = 2^r$ for some integer r, but no such r exists. ◆

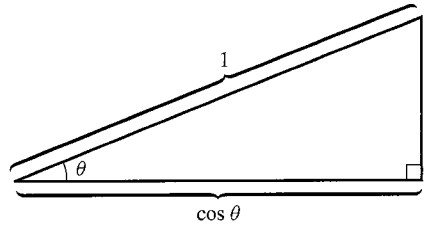
32.10 Theorem *Squaring the circle is impossible;* that is, given a circle, it is not always possible to construct with a straightedge and a compass a square having area equal to the area of the given circle.

Proof Let the given circle have a radius of 1, and hence an area of π. We would need to construct a square of side $\sqrt{\pi}$. But π is transcendental over $\mathbb{Q}$, so $\sqrt{\pi}$ is transcendental over $\mathbb{Q}$ also. ◆

32.11 Theorem *Trisecting the angle is impossible;* that is, there exists an angle that cannot be trisected with a straightedge and a compass.

Proof Figure 32.12 indicates that the angle θ can be constructed if and only if a segment of length $|\cos\theta|$ can be constructed. Now $60°$ is a constructible angle, and we shall show that it cannot be trisected. Note that

$$\begin{aligned} \cos 3\theta &= \cos(2\theta + \theta) \\ &= \cos 2\theta \cos\theta - \sin 2\theta \sin\theta \\ &= (2\cos^2\theta - 1)\cos\theta - 2\sin\theta\cos\theta\sin\theta \\ &= (2\cos^2\theta - 1)\cos\theta - 2\cos\theta(1 - \cos^2\theta) \\ &= 4\cos^3\theta - 3\cos\theta. \end{aligned}$$



32.12 Figure

[We realize that many students today have not seen the trigonometric identities we just used. Exercise 1 repeats Exercise 40 of Section 1 and asks you to prove the identity $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ from Euler's formula.]

Let $\theta = 20°$, so that $\cos 3\theta = \frac{1}{2}$, and let $\alpha = \cos 20°$. From the identity $4\cos^3\theta - 3\cos\theta = \cos 3\theta$, we see that

$$4\alpha^3 - 3\alpha = \frac{1}{2}.$$

Thus α is a zero of $8x^3 - 6x - 1$. This polynomial is irreducible in $\mathbb{Q}[x]$, since, by Theorem 23.11, it is enough to show that it does not factor in $\mathbb{Z}[x]$. But a factorization in $\mathbb{Z}[x]$ would entail a linear factor of the form $(8x \pm 1)$, $(4x \pm 1)$, $(2x \pm 1)$, or $(x \pm 1)$. We can quickly check that none of the numbers $\pm\frac{1}{8}, \pm\frac{1}{4}, \pm\frac{1}{2}$, and ± 1 is a zero of $8x^3 - 6x - 1$. Thus

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3,$$

so by Corollary 32.8, α is not constructible. Hence 60° cannot be trisected. ◆

■ Historical Note

Greek mathematicians as far back as the fourth century B.C. had tried without success to find geometric constructions using straightedge and compass to trisect the angle, double the cube, and square the circle. Although they were never able to prove that such constructions were impossible, they did manage to construct the solutions to these problems using other tools, including the conic sections.

It was Carl Gauss in the early nineteenth century who made a detailed study of constructibility in connection with his solution of cyclotomic equations, the equations of the form $x^p - 1 = 0$ with p prime whose roots form the vertices of a regular p-gon. He showed that although all such equations are solvable using radicals, if $p - 1$ is not a power of 2, then the solutions must involve roots higher than the second. In fact, Gauss asserted that anyone who attempted to find a geometric construction for a p-gon where $p - 1$ is not a power of 2 would "spend his time uselessly." Interestingly, Gauss did not prove the assertion that such constructions were impossible. That was accomplished in 1837 by Pierre Wantzel (1814–1848), who in fact proved Corollary 32.8 and also demonstrated Theorems 32.9 and 32.11. The proof of Theorem 32.10, on the other hand, requires a proof that π is transcendental, a result finally achieved in 1882 by Ferdinand Lindemann (1852–1939).

Note that the regular n-gon is constructible for $n \geq 3$ if and only if the angle $2\pi/n$ is constructible, which is the case if and only if a line segment of length $\cos(2\pi/n)$ is constructible.

EXERCISES 32

Computations

1. Prove the trigonometric identity $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ from the Euler formula, $e^{i\theta} = \cos\theta + i\sin\theta$.

Concepts

2. Mark each of the following true or false.

_____ **a.** It is impossible to double any cube of constructible edge by compass and straightedge constructions.

_____ **b.** It is impossible to double every cube of constructible edge by compass and straightedge constructions.

_____ **c.** It is impossible to square any circle of constructible radius by straightedge and compass constructions.

_____ **d.** No constructible angle can be trisected by straightedge and compass constructions.

_____ **e.** Every constructible number is of degree 2^r over $\mathbb{Q}$ for some integer $r \geq 0$.

_____ **f.** We have shown that every real number of degree 2^r over $\mathbb{Q}$ for some integer $r \geq 0$ is constructible.

_____ **g.** The fact that factorization of a positive integer into a product of primes is unique (up to order) was used strongly at the conclusion of Theorems 32.9 and 32.11.

_____ **h.** Counting arguments are exceedingly powerful mathematical tools.

_____ **i.** We can find any constructible number in a finite number of steps by starting with a given segment of unit length and using a straightedge and a compass.

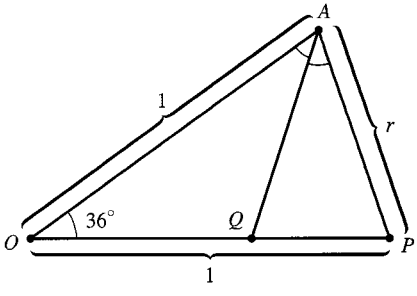
_____ **j.** We can find the totality of all constructible numbers in a finite number of steps by starting with a given segment of unit length and using a straightedge and a compass.

Theory

3. Using the proof of Theorem 32.11, show that the regular 9-gon is not constructible.

4. Show *algebraically* that it is possible to construct an angle of 30°.

5. Referring to Fig. 32.13, where $\overline{AQ}$ bisects angle OAP, show that the regular 10-gon is constructible (and therefore that the regular pentagon is also). [*Hint:* Triangle OAP is similar to triangle APQ. Show algebraically that r is constructible.]



32.13 Figure

In Exercises 6 through 9 use the results of Exercise 5 where needed to show that the statement is true.

6. The regular 20-gon is constructible.
7. The regular 30-gon is constructible.
8. The angle 72° can be trisected.
9. The regular 15-gon can be constructed.
10. Suppose you wanted to explain roughly in just three or four sentences, for a high school plane geometry teacher who never had a course in abstract algebra, how it can be shown that it is impossible to trisect an angle of 60°. Write down what you would say.

SECTION 33 FINITE FIELDS

The purpose of this section is to determine the structure of all finite fields. We shall show that for every prime p and positive integer n, there is exactly one finite field (up to isomorphism) of order p^n. This field $\mathrm{GF}(p^n)$ is usually referred to as the **Galois field of order** p^n. We shall be using quite a bit of our material on cyclic groups. The proofs are simple and elegant.

The Structure of a Finite Field

We now show that all finite fields must have prime-power order.

33.1 Theorem Let E be a finite extension of degree n over a finite field F. If F has q elements, then E has q^n elements.

Proof Let $\{\alpha_1, \cdots, \alpha_n\}$ be a basis for E as a vector space over F. By Exercise 21 of Section 30, every $\beta \in E$ can be *uniquely* written in the form

$$\beta = b_1\alpha_1 + \cdots + b_n\alpha_n$$

for $b_i \in F$. Since each b_i may be any of the q elements of F, the total number of such distinct linear combinations of the α_i is q^n. ◆

33.2 Corollary If E is a finite field of characteristic p, then E contains exactly p^n elements for some positive integer n.

Proof Every finite field E is a finite extension of a prime field isomorphic to the field $\mathbb{Z}_p$, where p is the characteristic of E. The corollary follows at once from Theorem 33.1. ◆

We now turn to the study of the multiplicative structure of a finite field. The following theorem will show us how any finite field can be formed from the prime subfield.

33.3 Theorem Let E be a field of p^n elements contained in an algebraic closure $\overline{\mathbb{Z}}_p$ of $\mathbb{Z}_p$. The elements of E are precisely the zeros in $\overline{\mathbb{Z}}_p$ of the polynomial $x^{p^n} - x$ in $\mathbb{Z}_p[x]$.

Proof The set E^* of nonzero elements of E forms a multiplicative group of order $p^n - 1$ under the field multiplication. For $\alpha \in E^*$, the order of α in this group divides the order $p^n - 1$ of the group. Thus for $\alpha \in E^*$, we have $\alpha^{p^n-1} = 1$, so $\alpha^{p^n} = \alpha$. Therefore, every element in E is a zero of $x^{p^n} - x$. Since $x^{p^n} - x$ can have at most p^n zeros, we see that E contains precisely the zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}}_p$. ◆

33.4 Definition An element α of a field is an ***n*th root of unity** if $\alpha^n = 1$. It is a **primitive *n*th root of unity** if $\alpha^n = 1$ and $\alpha^m \neq 1$ for $0 < m < n$. ■

Thus the nonzero elements of a finite field of p^n elements are all $(p^n - 1)$th roots of unity.

Recall that in Corollary 23.6, we showed that the multiplicative group of nonzero elements of a finite field is cyclic. This is a very important fact about finite fields; it has actually been applied to algebraic coding. For the sake of completeness in this section, we now state it here as a theorem, give a corollary, and illustrate with an example.

33.5 Theorem The multiplicative group $\langle F^*, \cdot \rangle$ of nonzero elements of a finite field F is cyclic.

Proof See Corollary 23.6. ◆

33.6 Corollary A finite extension E of a finite field F is a simple extension of F.

Proof Let α be a generator for the cyclic group E^* of nonzero elements of E. Then $E = F(\alpha)$ ◆

33.7 Example Consider the finite field $\mathbb{Z}_{11}$. By Theorem 33.5 $\langle \mathbb{Z}_{11}{}^*, \cdot \rangle$ is cyclic. Let us try to find a generator of $\mathbb{Z}_{11}{}^*$ by brute force and ignorance. We start by trying 2. Since $|\mathbb{Z}_{11}{}^*| = 10$, 2 must be an element of $\mathbb{Z}_{11}{}^*$ of order dividing 10, that is, either 2, 5, or 10. Now

$$2^2 = 4, \quad 2^4 = 4^2 = 5, \quad \text{and} \quad 2^5 = (2)(5) = 10 = -1.$$

Thus neither 2^2 nor 2^5 is 1, but, of course, $2^{10} = 1$, so 2 is a generator of $\mathbb{Z}_{11}{}^*$, that is, 2 is a primitive 10th root of unity in $\mathbb{Z}_{11}$. We were lucky.

By the theory of cyclic groups, all the generators of $Z_{11}{}^*$, that is, all the primitive 10th roots of unity in $\mathbb{Z}_{11}$, are of the form 2^n, where n is relatively prime to 10. These elements are

$$2^1 = 2, \quad 2^3 = 8, \quad 2^7 = 7, \quad 2^9 = 6.$$

Historical Note

Although Carl F. Gauss had shown that the set of residues modulo a prime p satisfied the field properties, it was Evariste Galois (1811–1832) who first dealt with what he called "incommensurable solutions" to the congruence $F(x) \equiv 0 \pmod{p}$, where $F(x)$ is an nth degree irreducible polynomial modulo p. He noted in a paper written in 1830 that one should consider the roots of this congruence as "a variety of imaginary symbols" that one can use in calculations just as one uses $\sqrt{-1}$. Galois then showed that if α is any solution of $F(x) \equiv 0 \pmod{p}$, the expression $a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1}$ takes on precisely p^n different values. Finally, he proved results equivalent to Theorems 33.3 and 33.5 of the text.

Galois' life was brief and tragic. He showed brilliance in mathematics early on, publishing several papers before he was 20 and essentially established the basic ideas of Galois theory. He was, however, active in French revolutionary politics following the July revolution of 1830. In May 1831, he was arrested for threatening the life of King Louis-Philippe. Though he was acquitted, he was rearrested for participating, heavily armed, in a republican demonstration on Bastille Day of that year. Two months after his release from prison the following March, he was killed in a duel, "the victim of an infamous coquette and her two dupes"; the previous night he had written a letter to a friend clarifying some of his work in the theory of equations and requesting that it be studied by other mathematicians. Not until 1846, however, were his major papers published; it is from that date that his work became influential.

The primitive 5th roots of unity in $\mathbb{Z}_{11}$ are of the form 2^m, where the gcd of m and 10 is 2, that is,

$$2^2 = 4, \quad 2^4 = 5, \quad 2^6 = 9, \quad 2^8 = 3.$$

The primitive square root of unity in $\mathbb{Z}_{11}$ is $2^5 = 10 = -1$. ▲

The Existence of GF(p^n)

We turn now to the question of the existence of a finite field of order p^r for every prime power p^r, $r > 0$. We need the following lemma.

33.8 Lemma If F is a field of prime characteristic p with algebraic closure $\bar{F}$, then $x^{p^n} - x$ has p^n distinct zeros in $\bar{F}$. ◆

Proof Because $\bar{F}$ is algebraically closed, $x^{p^n} - x$ factors over that field into a product of linear factors $x - \alpha$, so it suffices to show that none of these factors occurs more than once in the factorization.

Since we have not introduced an algebraic theory of derivatives, this elegant technique is not available to us, so we proceed by long division. Observe that 0 is a zero of $x^{p^n} - x$ of multiplicity 1. Suppose $\alpha \neq 0$ is a zero of $x^{p^n} - x$, and hence is a zero of $f(x) = x^{p^n-1} - 1$. Then $x - \alpha$ is a factor of $f(x)$ in $\bar{F}[x]$, and by long division, we find

that

$$\frac{f(x)}{(x-\alpha)} = g(x)$$
$$= x^{p^n-2} + \alpha x^{p^n-3} + \alpha^2 x^{p^n-4} + \cdots + \alpha^{p^n-3}x + \alpha^{p^n-2}.$$

Now $g(x)$ has $p^n - 1$ summands, and in $g(\alpha)$, each summand is

$$\alpha^{p^n-2} = \frac{\alpha^{p^n-1}}{\alpha} = \frac{1}{\alpha}.$$

Thus

$$g(\alpha) = [(p^n - 1) \cdot 1]\frac{1}{\alpha} = -\frac{1}{\alpha}.$$

since we are in a field of characteristic p. Therefore, $g(\alpha) \neq 0$, so α is a zero of $f(x)$ of multiplicity 1. ◆

33.9 Lemma If F is a field of prime characteristic p, then $(\alpha + \beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$ for all $\alpha, \beta \in F$ and all positive integers n. ◆

Proof Let $\alpha, \beta \in F$. Applying the binomial theorem to $(\alpha + \beta)^p$, we have

$$(\alpha + \beta)^p = \alpha^p + (p \cdot 1)\alpha^{p-1}\beta + \left(\frac{p(p-1)}{2} \cdot 1\right)\alpha^{p-2}\beta^2$$
$$+ \cdots + (p \cdot 1)\alpha\beta^{p-1} + \beta^p$$
$$= \alpha^p + 0\alpha^{p-1}\beta + 0\alpha^{p-2}\beta^2 + \cdots + 0\alpha\beta^{p-1} + \beta^p$$
$$= \alpha^p + \beta^p.$$

Proceeding by induction on n, suppose that we have $(\alpha + \beta)^{p^{n-1}} = \alpha^{p^{n-1}} + \beta^{p^{n-1}}$. Then $(\alpha + \beta)^{p^n} = [(\alpha + \beta)^{p^{n-1}}]^p = (\alpha^{p^{n-1}} + \beta^{p^{n-1}})^p = \alpha^{p^n} + \beta^{p^n}$. ◆

33.10 Theorem A finite field $\mathrm{GF}(p^n)$ of p^n elements exists for every prime power p^n.

Proof Let $\overline{\mathbb{Z}}_p$ be an algebraic closure of $\mathbb{Z}_p$, and let K be the subset of $\overline{\mathbb{Z}}_p$ consisting of all zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}}_p$. Let $\alpha, \beta \in K$. Lemma 33.9 shows that $(\alpha + \beta) \in K$, and the equation $(\alpha\beta)^{p^n} = \alpha^{p^n}\beta^{p^n} = \alpha\beta$ shows that $\alpha\beta \in K$. From $\alpha^{p^n} = \alpha$ we obtain $(-\alpha)^{p^n} = (-1)^{p^n}\alpha^{p^n} = (-1)^{p^n}\alpha$. If p is an odd prime, then $(-1)^{p^n} = -1$ and if $p = 2$ then $-1 = 1$. Thus $(-\alpha)^{p^n} = -\alpha$ so $-\alpha \in K$. Now 0 and 1 are zeros of $x^{p^n} - x$. For $\alpha \neq 0$, $\alpha^{p^n} = \alpha$ implies that $(1/\alpha)^{p^n} = 1/\alpha$. Thus K is a subfield of $\overline{\mathbb{Z}}_p$ containing $\mathbb{Z}_p$. Therefore, K is the desired field of p^n elements, since Lemma 33.8 showed that $x^{p^n} - x$ has p^n distinct zeros in $\overline{\mathbb{Z}}_p$. ◆

33.11 Corollary If F is any finite field, then for every positive integer n, there is an irreducible polynomial in $F[x]$ of degree n.

Proof Let F have $q = p^r$ elements, where p is the characteristic of F. By Theorem 33.10, there is a field $K \leq \bar{F}$ containing $\mathbb{Z}_p$ (up to isomorphism) and consisting precisely of the

zeros of $x^{p^{rn}} - x$. We want to show $F \leq K$. Every element of F is a zero of $x^{p^r} - x$, by Theorem 33.3. Now $p^{rs} = p^r p^{r(s-1)}$. Applying this equation repeatedly to the exponents and using the fact that for $\alpha \in F$ we have $\alpha^{p^r} = \alpha$, we see that for $\alpha \in F$,

$$\alpha^{p^{rn}} = \alpha^{p^{r(n-1)}} = \alpha^{p^{r(n-2)}} = \cdots = \alpha^{p^r} = \alpha.$$

Thus $F \leq K$. Then Theorem 33.1 shows that we must have $[K : F] = n$. We have seen that K is simple over F in Corollary 33.6 so $K = F(\beta)$ for some $\beta \in K$. Therefore, $\text{irr}(\beta, F)$ must be of degree n. ◆

33.12 Theorem Let p be a prime and let $n \in \mathbb{Z}^+$. If E and E' are fields of order p^n, then $E \simeq E'$.

Proof Both E and E' have $\mathbb{Z}_p$ as prime field, up to isomorphism. By Corollary 33.6, E is a simple extension of $\mathbb{Z}_p$ of degree n, so there exists an irreducible polynomial $f(x)$ of degree n in $\mathbb{Z}_p[x]$ such that $E \simeq \mathbb{Z}_p[x]/\langle f(x)\rangle$. Because the elements of E are zeros of $x^{p^n} - x$, we see that $f(x)$ is a factor of $x^{p^n} - x$ in $\mathbb{Z}_p[x]$. Because E' also consists of zeros of $x^{p^n} - x$, we see that E' also contains zeros of irreducible $f(x)$ in $\mathbb{Z}_p[x]$. Thus, because E' also contains exactly p^n elements, E' is also isomorphic to $\mathbb{Z}_p[x]/\langle f(x)\rangle$. ◆

Finite fields have been used in algebraic coding. In an article in the *American Mathematical Monthly* 77 (1970): 249–258, Norman Levinson constructs a linear code that can correct up to three errors using a finite field of order 16.

EXERCISES 33

Computations

In Exercises 1 through 3, determine whether there exists a finite field having the given number of elements. (A calculator may be useful.)

1. 4096 **2.** 3127 **3.** 68,921

4. Find the number of primitive 8th roots of unity in GF(9).

5. Find the number of primitive 18th roots of unity in GF(19).

6. Find the number of primitive 15th roots of unity in GF(31).

7. Find the number of primitive 10th roots of unity in GF(23).

Concepts

8. Mark each of the following true or false.

_____ **a.** The nonzero elements of every finite field form a cyclic group under multiplication.

_____ **b.** The elements of every finite field form a cyclic group under addition.

_____ **c.** The zeros in $\mathbb{C}$ of $(x^{28} - 1) \in \mathbb{Q}[x]$ form a cyclic group under multiplication.

_____ **d.** There exists a finite field of 60 elements.

_____ **e.** There exists a finite field of 125 elements.

_____ **f.** There exists a finite field of 36 elements.

_____ **g.** The complex number i is a primitive 4th root of unity.

_____ **h.** There exists an irreducible polynomial of degree 58 in $\mathbb{Z}_2[x]$.

_____ **i.** The nonzero elements of $\mathbb{Q}$ form a cyclic group $\mathbb{Q}^*$ under field multiplication.

_____ **j.** If F is a finite field, then every isomorphism mapping F onto a subfield of an algebraic closure $\bar{F}$ of F is an automorphism of F.

Theory

9. Let $\overline{\mathbb{Z}}_2$ be an algebraic closure of $\mathbb{Z}_2$, and let $\alpha, \beta \in \overline{\mathbb{Z}}_2$ be zeros of $x^3 + x^2 + 1$ and of $x^3 + x + 1$, respectively. Using the results of this section, show that $\mathbb{Z}_2(\alpha) = \mathbb{Z}_2(\beta)$.

10. Show that every irreducible polynomial in $\mathbb{Z}_p[x]$ is a divisor of $x^{p^n} - x$ for some n.

11. Let F be a finite field of p^n elements containing the prime subfield $\mathbb{Z}_p$. Show that if $\alpha \in F$ is a generator of the cyclic group $\langle F^*, \cdot \rangle$ of nonzero elements of F, then $\deg(\alpha, \mathbb{Z}_p) = n$.

12. Show that a finite field of p^n elements has exactly one subfield of p^m elements for each divisor m of n.

13. Show that $x^{p^n} - x$ is the product of all monic irreducible polynomials in $\mathbb{Z}_p[x]$ of a degree d dividing n.

14. Let p be an odd prime.

a. Show that for $a \in \mathbb{Z}$, where $a \not\equiv 0 \pmod{p}$, the congruence $x^2 \equiv a \pmod{p}$ has a solution in $\mathbb{Z}$ if and only if $a^{(p-1)/2} \equiv 1 \pmod{p}$. [*Hint:* Formulate an equivalent statement in the finite field $\mathbb{Z}_p$, and use the theory of cyclic groups.]

b. Using part (a), determine whether or not the polynomial $x^2 - 6$ is irreducible in $\mathbb{Z}_{17}[x]$.